

QUANTUM INFORMATION SCIENCE

Quantum Information Science

Information, computation, communication, and sensing
Developed syllabus and study text

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CyberColombia · Quantum Colombia Alliance

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This volume develops a complete QIS curriculum in sixteen grouped thematic chapters. Its 112 thematic sections contain explanatory prose, central results, assumptions, and detailed study scope. Worked examples, exercises, selected solutions, laboratory specifications, and teaching routes support the text. Complete advanced proofs, a full solution manual, and executable laboratory notebooks are beyond this curriculum edition.

Companion prerequisite: *Mathematical and Physical Foundations for Quantum Computing*, version 09.

Typeset with XeLaTeX using the same IBM Plex fonts, 17 by 24 cm geometry, color palette, cover artwork, chapter hierarchy, and pedagogical environments as the English foundations volume.

Purpose and scope

Quantum information science studies the storage, processing, transmission, protection, and extraction of information under quantum physical laws. This English volume follows *Mathematical and Physical Foundations for Quantum Computing*, version 09, and develops the next stage of the series in sixteen grouped thematic chapters.

The curriculum covers operational theory, information and resources, communication and security, computation and algorithms, physical systems, reliability, sensing, and thermodynamics. Related subjects are developed as sections within a coherent chapter instead of becoming separate short chapters. Laboratories belong to their thematic chapters; integrated projects are collected in an appendix.

Learning objectives

- Explain quantum information tasks through physical models, information measures, and explicit resource constraints.
- Develop and evaluate communication protocols, algorithms, protection schemes, and sensing strategies.
- Connect mathematical results to physical implementation, uncertainty, and reproducible evidence.
- Recognize the assumptions and limits of security, speedup, certification, and thermodynamic claims.

A developed curriculum

Each of the 112 thematic sections contains explanatory prose, a mathematical result or physical mechanism where appropriate, an argument or justification, and its domain of application. The detailed study scope identifies the subtopics to consolidate. Each chapter includes objectives, prerequisite references, a worked example, four assessment problems, a common pitfall, an integrated laboratory specification, and an APA-style reading list. Selected analytical solutions, notation, teaching routes, and six integrated projects complete the instructional structure.

The text develops the curriculum beyond short topic descriptions. It is not an exhaustive proof-level research monograph: advanced topics identify the additional derivations and specialized study needed. The sixteen laboratory activities are specifications with validation criteria; this edition does not claim to supply executable notebooks or a complete solution manual.

The boundary with Foundations

The prerequisite volume is assumed completed. Its topics are named and referenced for further study, without a second exposition, repeated elementary worked examples, or laboratories devoted to the same introductory task. Applying a known tool to a new QIS problem remains necessary: using a reduced state to analyze a channel is different from teaching partial trace again.

Targeted prerequisite references

Linear algebra and complex notation: Foundations, Chapters 1–2. Probability and sampling: Chapter 3. Hilbert spaces, operators, measurement, and density matrices: Chapters 4–7. Elementary circuits: Chapter 8. Hamiltonians and continuous systems: Chapters 9–11. Ising, QUBO, and introductory annealing: Chapter 12. Study guidance and environment setup: Chapters 13–14. For further study of these topics, return to those chapters; each QIS chapter supplies more precise section references.

Specialized tools absent from Foundations are introduced when needed: trace norms and channel representations in Chapter 1; entropy and coding in Chapter 3; singular-value and Schmidt decompositions in Chapter 4; complexity in Chapter 7; fermionic mappings in Chapter 9; Gaussian phase space in Chapter 11; classical and quantum error-correcting codes in Chapter 12; statistical estimation in Chapter 15; and thermal resource theory in Chapter 16. None is silently imposed as prior coursework.

The revision audit accompanying this edition records the disposition of all 34 chapters in version 01 and maps removed prerequisite material to the actual sections of Foundations. In particular, elementary Bell preparation, mixture calculations, ideal Bloch tomography, isolated two-level evolution, and basic QUBO conversion are not reused as new QIS lessons.

Organization and use

Part I develops operations, information, and resources. Part II addresses communication and cryptography. Part III treats computational models, algorithms, simulation, optimization, and learning. Part IV connects physical architectures with correction, fault tolerance, and characterization. Part V completes the coverage with metrology and thermodynamics.

Prior QIS chapters are listed at each opening, and the full sequence respects those dependencies. Advanced sections may be deferred in a survey, with cross-references retained for later specialization. The proposed four-course route in Appendix C allows sustained study; chapter count does not prescribe semester count. Appendix D supplies projects that connect the branches.

The visual and pedagogical design follows Foundations: the same typography, page geometry, colors, chapter hierarchy, learning boxes, examples, exercises, references, appendices, and index. The prerequisite book itself is unchanged.

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PART I

Operational theory and quantum information

1 Generalized measurements, channels, and distinguishability

Learning objectives

- Design a generalized measurement and its conditional update.
- Verify physical channel representations.
- Quantify the distinguishability of preparations and transformations.

Prior QIS chapters: None beyond Foundations.

Foundations reference

Positive operators, projective measurement, the Bloch sphere, density matrices, and partial trace. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 5.5, 6.1, 6.3, 7.3, 7.4. These prerequisite topics are not redeveloped here.

1.1 POVMs and quantum instruments

A detector with outcomes x is described by effects $E_x \geq 0$ satisfying $\sum_x E_x = I$. The probability rule is $p_x = \text{tr}(E_x \rho)$. Unlike a projective measurement, the effects need not be orthogonal or idempotent. This additional freedom describes noisy detectors,

measurements with more outcomes than the input dimension, and deliberate inconclusive decisions.

Disturbance requires extra information. An instrument has maps $\mathcal{I}_x(\rho) = \sum_a K_{xa} \rho K_{xa}^\dagger$, with $E_x = \sum_a K_{xa}^\dagger K_{xa}$. For $p_x > 0$, the conditional output is $\mathcal{I}_x(\rho)/p_x$; forgetting the outcome gives $\sum_x \mathcal{I}_x(\rho)$. Thus two devices can have identical effects and different output states. In sequential experiments this difference changes later observations, even when the first outcome statistics agree.

Detailed study scope. Develop positive operator-valued measures and outcome-conditioned completely positive maps. Separate outcome probabilities from disturbance, including sequential measurements, weak monitoring, and inconclusive outcomes. Connect an instrument to the channel obtained by discarding its classical outcome.

1.2 Measurement dilation and discrimination

Naimark dilation implements a POVM by introducing an ancilla, applying an isometry, and reading a projective outcome on a larger space. The extra outcomes are physical records, not extra distinguishable dimensions hidden in the original input. The construction also clarifies why a measurement design must specify how its record is processed into a decision.

For two hypotheses with priors q and $1-q$, let $\Delta = q\rho_0 - (1-q)\rho_1$. Assign the positive eigenspace of Δ to hypothesis zero. Maximizing the decision probability gives $P_{\text{succ}} = (1 + \|\Delta\|_1)/2$. Unambiguous discrimination instead forbids erroneous conclusive answers and permits an inconclusive outcome. These tasks optimize different losses. No cloning follows because preserving an input overlap s while producing two copies would require $s = s^2$, which excludes distinct nonorthogonal pure states.

Detailed study scope. Construct ancilla-assisted measurements through Naimark dilation. Introduce Bayesian decision costs, minimum-error discrimination, the Helstrom bound, and unambiguous discrimination. Derive the no-cloning obstruction for nonorthogonal inputs and explain the no-broadcasting extension.

1.3 Complete positivity and Kraus maps

A channel must remain positive when it acts on one part of any larger state. This requirement is complete positivity: $\text{id}_R \otimes \mathcal{E}$ must be positive for every reference R . In finite dimensions, a map is completely positive exactly when it admits a Kraus representation $\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger$. Trace preservation requires $\sum_a K_a^\dagger K_a = I$.

Transposition is positive on an isolated state but fails this extension test on entangled inputs. By contrast, every Kraus representation stays positive after adjoining a reference because each extended term is a positive congruence. A selected measurement branch obeys $\sum_a K_a^\dagger K_a \leq I$. Renormalizing that branch makes the conditional state assignment nonlinear; it must not be confused with an unconditional deterministic channel.

Detailed study scope. Distinguish positivity from complete positivity through a reference system. Develop operator-sum representations, composition, trace preservation, trace-nonincreasing postselection, and unitality. Use transposition to expose a map that fails the physical extension requirement.

1.4 Choi representation and dilation

With $|\Omega\rangle = \sum_i |i\rangle|i\rangle$, define $J(\mathcal{E}) = (\text{id} \otimes \mathcal{E})(|\Omega\rangle\langle\Omega|)$. Then $J \geq 0$ tests complete positivity and $\text{tr}_{\text{out}} J = I_{\text{in}}$ tests trace preservation. Channel action is recovered by $\mathcal{E}(\rho) = \text{tr}_{\text{in}}[(\rho^T \otimes I)J]$. The transpose and subsystem order are essential conventions.

A Kraus family constructs the isometry $V = \sum_a K_a \otimes |a\rangle_E$. Tracing out E gives the channel; tracing out the output gives a complementary channel. This identifies information transferred to an inaccessible environment. Different Kraus families may represent the same accessible process, so a Kraus label is not a unique microscopic event unless an environmental measurement model supplies that interpretation.

Detailed study scope. Fix an unnormalized Choi convention, derive its positivity and output-trace constraints, and reconstruct channel action. Introduce Stinespring isometries and complementary channels as descriptions of information leakage to an environment.

1.5 Noise channels and operational models

A useful noise model describes both the altered information and the available record. The dephasing family $(1 - p)\rho + pZ\rho Z$ multiplies off-diagonal entries by $1 - 2p$. Complete dephasing occurs at $p = 1/2$; at $p = 1$ the operation is a reversible phase flip. Depolarization may instead be parameterized by $\mathcal{D}_\lambda(\rho) = (1 - \lambda)\rho + \lambda I/d$ for $0 \leq \lambda \leq 1$, and should not inherit a different convention silently.

Amplitude damping transfers excitation to the environment and is generally nonunital. Erasure replaces the input by an orthogonal flag, revealing that a use failed without revealing the lost quantum information. These models lead to different decoding and capacity questions. Model selection therefore requires more than matching one decay curve: it must reproduce the observations and records relevant to the target task.

Detailed study scope. Compare dephasing, depolarization, amplitude damping, and erasure through the information tasks they affect. State parameter conventions and distinguish flagged erasure from unflagged errors. Identify when a fitted channel depends on preparation assumptions.

1.6 Distances and continuity

The trace norm is the sum of singular values. Trace distance $T(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$ is the largest probability difference achievable by one measurement event; this makes it a task-based error measure. A channel cannot increase it, because any output measurement can be pulled back to a valid input measurement.

Use the squared-fidelity convention $F(\rho, \sigma) = (\text{tr } \sqrt{\sqrt{\rho}\sigma\sqrt{\rho}})^2$. It obeys $1 - \sqrt{F} \leq T \leq \sqrt{1 - F}$. For a pure target, fidelity is its projector expectation, whereas trace distance bounds all event probabilities. A high fidelity can therefore support a quantitative operational statement through these inequalities, but the statement depends on whether the target is a state, a process, or an average over inputs.

Detailed study scope. Introduce trace norm, trace distance, squared Uhlmann fidelity, and their bounds. Explain contractivity under physical processing and the difference between overlap with a target and an operational decision advantage.

1.7 Channel discrimination and reference systems [advanced]

For equal prior probabilities of two channels, discrimination succeeds with probability $1/2 + \|\mathcal{E} - \mathcal{F}\|_{\diamond}/4$. The diamond norm optimizes over an input entangled with a reference, allowing information stored in correlations to contribute to the test. In finite dimensions, a reference as large as the input suffices for the optimization.

Average fidelity probes an input average; diamond distance addresses the most distinguishable allowed experiment. Coherent errors can behave differently under these measures and can accumulate systematically across a circuit. A channel-level guarantee should specify the input space, treatment of leakage, and whether the classical success record is part of the output. Discarding failures changes the experiment and requires reporting success probability alongside conditional quality.

Detailed study scope. Develop ancilla-assisted discrimination and diamond distance. Contrast average gate fidelity with a worst-case channel guarantee; discuss why postselection changes the relevant success probability and cannot be treated as deterministic processing.

Example 1.1 · Minimum-error discrimination

For equally likely $|0\rangle$ and $|+\rangle$, the trace distance is $1/\sqrt{2}$. The Helstrom success probability is $P_{\text{succ}} = (1 + 1/\sqrt{2})/2 \simeq 0.8536$. Measuring only in the computational basis gives 0.75 under the best outcome decision rule. Optimizing the measurement therefore improves the task without revealing an arbitrary state vector.

Common pitfall

A POVM does not uniquely determine the state after measurement. An average fidelity also does not specify a worst-case discrimination error.

1.8 Exercises

Exercise 1.1

Analytical. Calculate the minimum error in the worked example and compare it with computational-basis readout.

Exercise 1.2

Conceptual. Explain why complete positivity tests inputs entangled with a reference.

Exercise 1.3

Diagnostic. Diagnose an erasure model that discards failed trials but reports unconditional success.

Exercise 1.4

Computational. Construct the Helstrom measurement numerically and test its decision rule on held-out simulated outcomes.

The analytical exercise is discussed in Appendix [A](#).

1.9 Laboratory: Discrimination and physical channels

Construct a two-state Helstrom detector and a qubit amplitude-damping channel. Validate detector positivity and completeness, and channel Choi positivity and trace preservation to 10^{-10} in double precision. Compare empirical discrimination error with its analytical value using a binomial interval. Deliver the

detector, channel convention, validation residuals, and an error-versus-sample plot; use at most two qubits.

Use the submission and validation contract in Appendix C.

References and further reading

Watrous, J. (2018). *The theory of quantum information*. Cambridge University Press. <https://jhwatrous.github.io/TQI.pdf>

Nielsen, M. A., & Chuang, I. L. (2010). *Quantum computation and quantum information* (10th anniversary ed.). Cambridge University Press.

2 Open-system dynamics and quantum control

Learning objectives

- Model relaxation and dephasing with explicit assumptions.
- Connect conditional trajectories to ensemble dynamics.
- Design and assess control under noise and physical constraints.

Prior QIS chapters: 1.

Foundations reference

Commutators, Hamiltonian evolution, spin dynamics, and driven two-level systems. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 5.6, 10.1, 10.3, 10.4, 10.7. These prerequisite topics are not redeveloped here.

2.1 Reduced dynamics and model assumptions

A reduced dynamical map is obtained from a specified joint preparation, an interaction, and environmental tracing. If every allowed input is prepared independently of the same environmental state, this construction defines a channel on all inputs. If preparation changes the environment or leaves input-dependent correlations, a single input-only channel may fail to describe the experiment.

This distinction matters in repeated control cycles. Apparent memory can originate in a bath, a slowly drifting control setting, or a preparation procedure. The first modeling step is to identify what is reset between trials and what persists. Comparing sequences with different delays or preparation histories can distinguish models that agree on one-time observations. A master equation is then a justified reduced description, not a replacement for these preparation assumptions.

Detailed study scope. Develop dynamical maps with specified system–environment preparation. Identify when initial correlations obstruct a single channel on arbitrary inputs, and distinguish environmental memory from preparation dependence or instrumental drift.

2.2 Lindblad generators

For a time-homogeneous Markovian model, the generator has the form

$$\dot{\rho} = -i[H, \rho] + \sum_a \gamma_a \left(L_a \rho L_a^\dagger - \frac{1}{2} \{L_a^\dagger L_a, \rho\} \right),$$

with $\hbar = 1$ and nonnegative rates. The anticommutator terms balance the jump terms so that trace is preserved. This structure generates completely positive trace-preserving evolution and makes the stationary-state condition $\mathcal{L}(\rho_*) = 0$ an explicit linear problem.

A microscopic weak-coupling derivation separates bath correlation time, relaxation time, and oscillation frequencies. Markov and secular approximations discard different effects;

neither follows merely from observing weak noise. Degenerate levels, strong driving, or structured environments can challenge the approximations. Numerical work should state whether the generator is phenomenological or derived from a specified coupling model.

Detailed study scope. Introduce the Gorini–Kossakowski–Sudarshan–Lindblad form for completely positive semigroup dynamics. Separate coherent terms, jump operators, rates, and steady states. Discuss the approximations behind weak-coupling and secular models rather than treating every environment as Markovian.

2.3 Relaxation, dephasing, and leakage

In the standard independent relaxation and pure-dephasing model, an off-diagonal element decays as $e^{-t/(2T_1)}e^{-t/T_\phi}$, yielding $1/T_2 = 1/(2T_1) + 1/T_\phi$. Consequently $T_2 \leq 2T_1$ in this model. A fitted violation calls for examining the model, estimation uncertainty, or inconsistent experimental conditions.

Finite-temperature dynamics has upward and downward rates; the stationary excited population need not vanish. Leakage introduces states outside the computational subspace, so a normalized qubit-only fit can hide lost population. Characterization should distinguish total survival, conditional computational-state quality, and return from leakage. Reporting only a single coherence time can conceal mechanisms that respond very differently to control and error correction.

Detailed study scope. Relate population relaxation, coherence decay, and pure dephasing in the standard independent-noise model. Include finite-temperature excitation, leakage out of the computational subspace, and the limits of fitting a single coherence time.

2.4 Monitored trajectories and feedback

A monitored environment yields a random measurement record and a conditional state. In a jump description, a detected event applies the appropriate jump operator and renormalizes;

between events, the no-click evolution is also conditional. Averaging records with their physical probabilities recovers the unconditional master equation.

Different environmental measurements can produce different unravelings of the same average dynamics. A diffusive record, for example, represents continuous noisy information rather than discrete counts. Feedback conditions later control on this record and introduces latency, detector efficiency, and classical-processing constraints. Coherent feedback instead keeps part of the controller quantum. The useful comparison is not whether trajectories look smooth, but whether the accessible record improves a specified control or estimation task.

Detailed study scope. Develop quantum-jump and diffusive descriptions conditioned on a measurement record. Connect trajectory averages with a master equation and distinguish measurement-based feedback from coherent feedback and unconditional control.

2.5 Noise spectroscopy and dynamical decoupling

For a dephasing interaction proportional to $\beta(t)Z$, ideal control pulses can change the sign of the accumulated noise through a switching function $y(t)$. The stochastic phase depends on $\int y(t)\beta(t) dt$. In stationary Gaussian models its variance can be written as a spectral integral involving the noise power spectrum and the sequence's filter function.

An echo cancels a static detuning because positive and negative evolution intervals contribute opposite phases. It does not cancel every fluctuating process: noise at frequencies transmitted by the filter remains. Finite pulse duration, amplitude errors, and pulse-induced leakage also add costs. Sequence design therefore balances spectral suppression against implementation error, and a noise-spectrum inference must specify the assumptions used to convert measured decay into spectral power.

Detailed study scope. Introduce filter functions, echo sequences, pulse imperfections, and spectral selectivity. Explain which noise components a sequence suppresses and when added control amplifies errors or invalidates a stationary-noise model.

2.6 Optimal and robust control

Write the controlled Hamiltonian as $H(t) = H_0 + \sum_k u_k(t)H_k$. Pulse optimization minimizes a chosen error functional over allowed controls $u_k(t)$, subject to amplitude, bandwidth, duration, and leakage constraints. The objective may be state transfer, a target channel, or a worst-case performance over uncertain parameters; these objectives are not interchangeable.

A nominal optimum can exploit details of an inaccurate model. Robust optimization instead evaluates an ensemble of detunings, coupling errors, or noise realizations. Gradient methods accelerate the search, but validation must use unseen parameter values and physical constraints. A pulse with lower simulated infidelity yet larger bandwidth than the control chain supports is not an improved implementation. Controllability establishes accessible operations in an ideal model, not their practical cost.

Detailed study scope. Formulate pulse design through an objective, amplitude and bandwidth bounds, and a noise model. Introduce gradient-based optimization, robustness ensembles, controllability, calibration, and validation against model mismatch.

2.7 Memory and numerical validation [advanced]

Complete-positive divisibility asks whether evolution between intermediate times can itself be represented by a channel. Information-backflow tests instead examine increases in distinguishability for selected state pairs. Their relation depends on the precise diagnostic and allowed auxiliary systems; a failed witness does not establish absence of memory.

Numerical validation must separate solver error from model error. Trace and Hermiticity residuals test algebraic consistency, while positivity checks whether the computed operator remains physical. Halving time steps tests convergence; comparison with a solvable limit tests implementation; trajectory averages require statistical uncertainty. Passing all these checks supports the simulated model, but experimental evidence is still required to justify that the model describes a particular device.

Detailed study scope. Compare complete-positive divisibility and information-backflow diagnostics without equating them universally. Check integrator convergence, physicality, trajectory uncertainty, and control robustness before interpreting a simulated improvement.

Example 2.1 · Separating relaxation from pure dephasing

In the independent relaxation and pure-dephasing model, $1/T_2 = 1/(2T_1) + 1/T_\phi$. For $T_1 = 40 \mu\text{s}$ and $T_\phi = 80 \mu\text{s}$, both contributions are $1/(80 \mu\text{s})$, so $T_2 = 40 \mu\text{s}$. This identifies two dissipative contributions; it is not a derivation of isolated spin precession.

Common pitfall

A nonexponential signal is not, by itself, a unique diagnosis of environmental memory. Drift and model mismatch can produce similar observations.

2.8 Exercises

Exercise 2.1

Analytical. With $T_1 = 60 \mu\text{s}$ and $T_2 = 80 \mu\text{s}$, infer T_ϕ in the stated model.

Exercise 2.2

Conceptual. Explain how one monitored trajectory differs from an ensemble prediction.

Exercise 2.3

Diagnostic. Assess a fit with $T_2 > 2T_1$ under the same independent-noise assumptions.

Exercise 2.4

Computational. Compare echo and free evolution under a declared quasistatic detuning distribution, including a pulse-error sweep.

The analytical exercise is discussed in Appendix A.

2.9 Laboratory: Dissipation and robust pulses

Integrate a driven dissipative qubit and compare a nominal pulse with a pulse optimized over detuning offsets. Report final channel performance across unseen offsets, not only the training objective. Validate trace and Hermiticity to 10^{-9} , test positivity within integration tolerance, and halve the time step. Deliver relaxation fits, a robustness plot, and evidence that the control bounds were respected.

Use the submission and validation contract in Appendix C.

References and further reading

Breuer, H.-P., & Petruccione, F. (2002). *The theory of open quantum systems*. Oxford University Press.

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3 Quantum entropy, coding, and communication limits

Learning objectives

- Interpret information measures through operational tasks.
- Distinguish source compression from channel coding.
- State communication rates with their assistance and asymptotic assumptions.

Prior QIS chapters: 1.

Foundations reference

Probability distributions, conditional probability, density matrices, and partial trace. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 3.1, 3.2, 3.3, 7.3, 7.4. These prerequisite topics are not redeveloped here.

3.1 Classical information and source coding

Shannon entropy $H(X) = -\sum_x p_x \log_2 p_x$ measures the asymptotic description length of an independent memoryless source. A typical length- n sequence has probability close to $2^{-nH(X)}$, while approximately $2^{nH(X)}$ such sequences carry nearly all

probability. Indexing this set explains the compression rate rather than merely assigning a formula to uncertainty.

Conditional entropy quantifies the remaining description cost when side information is available; mutual information $I(X : Y) = H(X) - H(X | Y)$ measures the reduction. Relative entropy compares two models and penalizes using a mismatched code. Finite blocks require a tolerance for atypical events and an explicit overhead. Compression is therefore specified by both storage length and failure probability, not by entropy alone.

Detailed study scope. Introduce Shannon entropy, conditional entropy, mutual information, and relative entropy as new tools. Develop typical sequences, lossless compression, and finite-block overhead, connecting probabilities already known from Foundations to communication costs.

3.2 Quantum entropy and compression

For a quantum source with average state ρ , the entropy $S(\rho) = -\text{tr}(\rho \log_2 \rho)$ determines the memoryless compression rate in qubits per signal. The spectral typical subspace of $\rho^{\otimes n}$ has dimension roughly $2^{nS(\rho)}$ and captures probability approaching one. Compression preserves this subspace coherently rather than measuring which eigenstring occurred.

This coherence requirement distinguishes quantum compression from saving classical preparation labels. A useful correctness criterion is preservation of correlations with a purifying reference, expressed through entanglement fidelity. Measuring and storing only an eigenlabel can reproduce some source statistics while destroying those correlations. At finite block size, one must specify the chosen subspace, retained probability, encoding and decoding operations, and how the failure component is handled.

Detailed study scope. Define von Neumann entropy through the spectrum and introduce typical subspaces and Schumacher compression for memoryless quantum sources. Explain why preserving a purification is stronger than reproducing a list of measurement probabilities.

3.3 Conditional information and correlations

Quantum conditional entropy is $S(A | B) = S(AB) - S(B)$; it can be negative because conditioning on a quantum memory differs from conditioning on a classical record. For a pure entangled state of AB , the joint entropy vanishes while the marginal entropy is positive. This negative value enters operational tasks such as state merging rather than representing a negative count of unknown classical outcomes.

Mutual information $I(A : B) = S(A) + S(B) - S(AB)$ measures total correlations, including both classical and quantum contributions. Strong subadditivity states $S(A | BC) \leq S(A | B)$ and underlies many processing inequalities. For a classical label X correlated with a quantum system, the block structure of the joint state connects these quantities to ensembles without making the preparation label directly observable on the quantum subsystem.

Detailed study scope. Develop subadditivity, strong subadditivity, mutual information, and negative quantum conditional entropy. Interpret entropy with a quantum memory and distinguish a classical label from inaccessible ensemble information.

3.4 Data processing and recoverability

Quantum relative entropy is $D(\rho \| \sigma) = \text{tr}[\rho(\log_2 \rho - \log_2 \sigma)]$ when the support of ρ lies within that of σ , and is infinite otherwise. Data processing gives $D(\mathcal{E}(\rho) \| \mathcal{E}(\sigma)) \leq D(\rho \| \sigma)$. An allowed physical transformation cannot improve the asymptotic distinguishability of two hypotheses merely by processing the same input.

Equality has a stronger meaning than numerical coincidence: under the appropriate support conditions, a recovery map can recover both states from their processed versions. Approximate recovery asks what a small information loss guarantees. These ideas connect statistical distinguishability, compression, and reversibility. They also warn against inferring global reversibility from preservation of one observable or one selected input state.

Detailed study scope. Establish the information loss expressed by relative-entropy data processing. Introduce hypothesis testing, support conditions, and recovery maps; identify the equality and approximation questions that connect information measures to reversible processing.

3.5 Accessible information and classical transmission

For an ensemble $\{p_x, \rho_x\}$, the Holevo quantity is $\chi = S(\sum_x p_x \rho_x) - \sum_x p_x S(\rho_x)$. It bounds the mutual information obtainable by measuring one encoded label. The bound is not a promise that a particular detector attains it, and ensembles with noncommuting signals need not reveal all their classical preparation information.

Channel coding adds block encoders, a message set, and a decoding error criterion. Collective decoding can exploit many channel uses jointly. General classical capacity requires optimization and, in general, regularization across blocks; a one-use optimized Holevo quantity is not universally additive. An achievable rate and a converse bound answer complementary questions: a construction shows what can be sent, while the converse excludes reliable transmission beyond the specified limit.

Detailed study scope. Derive the role of the Holevo quantity and distinguish a bound for one ensemble from an optimized coding theorem. Treat collective decoding, memoryless channel assumptions, and regularization in classical capacity.

3.6 Quantum, private, and assisted capacities

Quantum capacity concerns faithful transmission of unknown quantum information, equivalently preservation of entanglement with a reference. Coherent information $I_c = S(B) - S(E)$ for a channel dilation is central to achievable rates, but the general unassisted capacity involves a regularized optimization. Private classical capacity instead requires reliability for the receiver and secrecy from the complementary output.

With unlimited preshared entanglement, classical capacity has a single-letter expression through optimized input-reference mutual information. This assistance is a consumed or supplied resource and must appear in the task statement. The erasure channel makes the distinctions tangible: knowing which uses failed supports classical communication above the parameter range with positive unassisted quantum capacity. A capacity comparison is meaningful only when channel uses, direction of communication, and auxiliary assistance are fixed.

Detailed study scope. Introduce coherent information, quantum capacity, private classical capacity, and entanglement-assisted classical capacity. Contrast achievability with converse bounds, and flag channels for which a single-letter expression does not determine the general capacity.

3.7 Finite-block and one-shot information [advanced]

Asymptotic entropy suppresses finite-block details that matter when only a limited number of signals are available. Min-entropy connects to guessing probability; for a classical variable with quantum side information, $H_{\min}(X | B) = -\log_2 p_{\text{guess}}(X | B)$. Smoothing allows a controlled approximation to the actual state, introducing an explicit error parameter.

One-shot coding statements use these quantities to bound extractable randomness, compression length, or hypothesis-testing performance for a single finite instance. The asymptotic equipartition principle explains their approach to ordinary conditional entropy under independent repetition. It does not eliminate finite-size penalties in an actual experiment. Every finite-key or finite-block claim should retain its failure probability, smoothing convention, data dependence, and the assumptions under which the bound applies.

Detailed study scope. Introduce smooth min- and max-entropies, hypothesis-testing quantities, error parameters, and finite-block penalties. Explain their roles in cryptography and compression without replacing finite-size guarantees by asymptotic rates.

Example 3.1 · The flagged qubit erasure channel

For independent erasure probability p , the unassisted classical capacity is $C = 1 - p$ bits per channel use, and the unassisted quantum capacity is $Q = \max(0, 1 - 2p)$ qubits per use. At $p = 0.2$, these are 0.8 and 0.6. The receiver knows which uses were erased; a depolarizing channel does not provide that flag.

Common pitfall

A capacity is tied to a channel model and an assistance convention. A one-use information quantity is not automatically the capacity of a general channel.

3.8 Exercises

Exercise 3.1

Analytical. Evaluate C and Q for the erasure channel at $p = 0.4$ and $p = 0.6$.

Exercise 3.2

Conceptual. Explain the operational difference between Schumacher compression and transmitting through noise.

Exercise 3.3

Diagnostic. Diagnose a claim that positive one-use mutual information guarantees positive unassisted quantum capacity.

Exercise 3.4

Computational. Plot erasure capacities and estimate finite-block fluctuations in the number of successful transmissions.

The analytical exercise is discussed in Appendix A.

3.9 Laboratory: Compression and finite-block rates

For a two-eigenvalue quantum source, enumerate typical eigenvalue strings for blocks of at most twelve signals and compare retained probability with compression dimension. Separately simulate flagged erasures and report finite-block rate fluctuations. Validate enumeration by total probability and compare with entropy predictions as block size changes. Deliver rate-error curves and an explanation of source-coding versus channel-coding costs.

Use the submission and validation contract in Appendix C.

References and further reading

- Wilde, M. M. (2017). *Quantum information theory* (2nd ed.). Cambridge University Press. <https://www.markwilde.com/qit-notes.pdf>
- Watrous, J. (2018). *The theory of quantum information*. Cambridge University Press. <https://jhwatrous.github.io/TQI.pdf>

4 Entanglement, nonlocality, and quantum resources

Learning objectives

- Quantify and manipulate entanglement beyond elementary two-qubit tests.
- Distinguish entanglement, steering, nonlocality, and contextuality.
- Specify a resource theory and evaluate allowed conversions.

Prior QIS chapters: 1, 3.

Foundations reference

Spectral methods, tensor products, Bell states, elementary pure-state separability, and reduced states. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 1.7, 1.8, 4.5, 7.1, 7.2, 7.4, 8.2. These prerequisite topics are not redeveloped here.

4.1 Schmidt structure and purification

For a bipartite pure state with coefficient matrix C , singular-value decomposition writes $C = U\Sigma V^\dagger$ with nonnegative diagonal entries s_i . Substituting the corresponding local basis

changes gives the Schmidt form $\sum_i s_i |a_i\rangle |b_i\rangle$. The squared coefficients sum to one and are the nonzero eigenvalues of both reduced states. Thus entanglement entropy is the Shannon entropy of the Schmidt spectrum.

The same construction purifies a mixed state by assigning each eigenvalue to a correlated reference basis vector. Purifications with sufficiently large reference spaces differ by a reference isometry. This freedom matters when comparing implementations that have the same accessible state but different environmental descriptions. The general decomposition extends the elementary two-qubit rank test without repeating its derivation or the construction of a Bell pair.

Detailed study scope. Introduce singular-value decomposition locally and use it to obtain Schmidt coefficients in general bipartite dimensions. Develop purification freedom, entanglement entropy, and operational consequences of the reduced spectrum. Singular-value decomposition is new material here, not an entry prerequisite.

4.2 Mixed-state entanglement and witnesses

A mixed state is separable when it is a convex sum $\sum_j p_j \rho_j^A \otimes \rho_j^B$. Classical correlation is allowed in this definition; being nonproduct does not establish entanglement. Partial transposition preserves positivity of every such sum, yielding a necessary separability test. In dimensions 2×2 and 2×3 it is also sufficient, but this sufficiency fails in general.

An entanglement witness is a Hermitian observable whose expectation is nonnegative on all separable states but negative on at least one entangled state. It can require far fewer measurements than full reconstruction, at the price of detecting only some entangled states. Negativity sums the absolute negative eigenvalues of the partial transpose. Finite-data witness estimates need uncertainty intervals: a negative point estimate without adequate significance is not a certification.

Detailed study scope. Develop separable mixtures, partial transposition, negativity, concurrence for two qubits, and witness estimation. State when the positive-partial-transpose criterion is sufficient and why positive partial transpose can coexist with entanglement in larger systems.

4.3 LOCC and entanglement conversion

Local operations and classical communication permit each party to manipulate its own subsystem and coordinate through classical messages. They cannot create entanglement from separable inputs. For deterministic conversion of bipartite pure states, the input Schmidt-probability vector must be majorized by the output vector. In this convention a maximally entangled state can be converted to a product state, which provides a useful direction check.

Concentration distills nearly maximally entangled pairs from many weakly entangled pure states; dilution reverses the asymptotic pure-state task. For mixed states, entanglement cost and distillable entanglement need not coincide. Probabilistic conversion, catalytic assistance, and asymptotic approximation change the allowed transformation. A claimed conversion therefore needs an operation class, success probability, target accuracy, and accounting for any borrowed resource.

Detailed study scope. Define local operations and classical communication, majorization, concentration, dilution, and distillation. Contrast entanglement cost with distillable entanglement, and discuss bound entanglement and the importance of single-copy versus asymptotic conversion.

4.4 Multipartite entanglement

Multipartite entanglement has several inequivalent structures because different partitions and tasks expose different correlations. Losing one member of a GHZ resource can destroy the useful coherence, whereas a W resource can retain pairwise entanglement in a reduced state. Neither observation alone makes one family universally superior.

Graph states organize correlations through a graph and support measurement-based processing. Their usefulness depends on connectivity, allowed measurements, loss, and feed-forward. Genuine multipartite entanglement excludes mixtures that are separable across different bipartitions, a stronger requirement than entanglement across one fixed cut. Monogamy inequalities constrain how selected measures can be shared but depend on the measure and system dimensions. Characterization should therefore identify the target task and relevant partitions before choosing a witness or an entanglement measure.

Detailed study scope. Compare GHZ, W, and graph-state resources across cuts, losses, and measurement tasks. Introduce monogamy and genuine multipartite entanglement without assigning every task to a single scalar measure.

4.5 Bell nonlocality and steering

In a local hidden-variable model, conditional outcomes factorize given a shared variable and the settings are independent of that variable. For binary outcomes, deterministic assignments give $|A_0(B_0 + B_1) + A_1(B_0 - B_1)| = 2$; averaging yields the CHSH bound $|S| \leq 2$. Suitable entangled measurements reach $2\sqrt{2}$, while no-signalling alone allows a larger set of correlations.

Entanglement witnesses trust both measurement models; steering trusts one side; Bell tests can avoid trusting either measurement description under additional causal assumptions. Detection selection, setting dependence, communication, and memory can alter the inference. Statistical evidence must use a trial definition and an analysis compatible with those assumptions. A quantum correlation calculation predicts an ideal value; it does not by itself establish a loophole-controlled experimental test.

Detailed study scope. Derive the CHSH local bound and quantum violation under explicit causal and measurement assumptions. Distinguish trusted, one-sided trusted, and device-independent tests; analyze detection, locality, setting independence, memory, and finite-data significance.

4.6 Contextuality and reference frames

Contextuality asks whether outcomes can be assigned consistently to measurements independently of the compatible context in which they appear. It does not require spatially separated parties. A finite observable construction can expose incompatible value assignments even though each compatible set has a valid joint measurement model. The operational meaning depends on the compatibility and equivalence assumptions tested in the experiment.

A missing reference frame creates a different restriction. If relative phase is unknown, averaging over phase rotations produces a twirled description $\mathcal{G}(\rho) = \int dg U_g \rho U_g^\dagger$. Coherences between symmetry sectors may become inaccessible. Supplying a reference is therefore an operational resource. One must distinguish lack of a reference from actual physical dephasing, even when restricted observations give the same effective description.

Detailed study scope. Introduce noncontextual models for compatible measurements and distinguish contextuality from spatial nonlocality. Develop symmetry restrictions, twirling, and the operational consequences of missing phase or orientation references.

4.7 Coherence, asymmetry, and resource conversion [advanced]

A resource theory begins with free states and free operations, then asks which state transformations or tasks become possible with a resource. A monotone cannot increase under the chosen free operations, providing an obstruction to conversion. Different definitions of free coherence operations lead to different operational theories even when they share the same diagonal free states.

Relative entropy of coherence is $C_r(\rho) = S(\Delta\rho) - S(\rho)$ for dephasing Δ in a specified basis. Asymmetry instead depends on a symmetry action and available references. Magic describes resources beyond stabilizer processing and becomes a hardware

cost in fault tolerance. These examples show why the phrase “more quantum” is insufficient: an entangled, coherent, or non-stabilizer state is useful only relative to an allowed task and its resource account.

Detailed study scope. Specify free states, free operations, monotones, and conversion tasks. Compare coherence and asymmetry with entanglement, introduce catalysis and magic as resource concepts, and reserve stabilizer implementation and magic-state factories for Chapters 12–13.

Example 4.1 · A noisy entangled resource

Let $\rho_p = p|\Phi^+\rangle\langle\Phi^+| + (1-p)I/4$, where $0 \leq p \leq 1$. Its partial transpose has three eigenvalues $(1+p)/4$ and one $(1-3p)/4$. It is entangled exactly when $p > 1/3$, using sufficiency of the PPT test in two-qubit systems. The Bell state is assumed known; the new result is the mixed-state noise threshold.

Common pitfall

Failure to violate one Bell inequality does not prove separability. A positive partial transpose is not a universal separability certificate.

4.8 Exercises

Exercise 4.1

Analytical. Find the negativity of the example at $p = 0.6$ and at $p = 0.2$.

Exercise 4.2

Conceptual. Explain why LOCC conversion depends on the allowed resources and success convention.

Exercise 4.3

Diagnostic. Assess a claim of device independence based on trusted detector calibration and postselected detections.

Exercise 4.4

Computational. Compare the PPT threshold with the standard optimized CHSH value $2\sqrt{2}p$ for the same state family.

The analytical exercise is discussed in Appendix A.

4.9 Laboratory: Witnesses and Bell evidence

Sweep the noisy Bell-resource parameter and compare negativity, an entanglement witness, and CHSH correlators. Generate finite measurement records without discarding detector failures; state the chosen failure outcome convention. Validate the exact PPT and CHSH curves and report uncertainty for the sampled witness. Deliver the distinct entanglement and nonlocality thresholds, with the assumptions needed for each claim. Use the submission and validation contract in Appendix C.

References and further reading

- Horodecki, R., Horodecki, P., Horodecki, M., & Horodecki, K. (2009). Quantum entanglement. *Reviews of Modern Physics*, 81(2), 865–942. <https://doi.org/10.1103/RevModPhys.81.865>
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PART II

Quantum communication

5 Quantum communication protocols and networks

Learning objectives

- Account for quantum and classical resources in communication protocols.
- Analyze entanglement distribution with loss and memory.
- Evaluate a network by rate, fidelity, latency, and trust.

Prior QIS chapters: 1, 3, 4.

Foundations reference

Composite registers, reduced states, and controlled operations. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 7.1, 7.4, 8.2. These prerequisite topics are not redeveloped here.

5.1 Teleportation, dense coding, and remote operations

Teleportation transfers an unknown input using a shared entangled pair, a joint measurement, and a two-bit classical outcome. Conditional on that outcome, the receiver's state differs from the input by a known Pauli correction. Before receiving the message, averaging the possible corrections removes access to the

input information; this is why the protocol does not signal instantaneously.

Dense coding reverses the resource balance: a transmitted qubit and a preshared pair carry two classical bits through an appropriate joint decoding measurement. Remote state preparation assumes classical knowledge of the target and is a different task. Gate teleportation combines measurement with a prepared resource so that corrections implement or accompany a desired operation. In every case, the resource ledger must include entanglement generation, transmitted systems, classical messages, and conditional failures.

Detailed study scope. Derive teleportation through conditional corrections and account for the shared entangled pair and classical message. Compare dense coding, remote state preparation with stated knowledge, and gate teleportation. Track resources per successful use and prevent a no-signalling violation.

5.2 Entanglement swapping and purification

Entanglement swapping measures the two local halves of separate pairs at an intermediate node. Conditional corrections leave the remote halves entangled even though they did not interact directly. The input pairs, measurement outcome, correction rule, and success condition define the operation; imperfect pair preparation and local measurements both affect the final resource.

Purification consumes several imperfect pairs and uses local operations with classical coordination to retain fewer, better pairs. It improves conditional quality only in an appropriate input and gate-error regime, and its success probability reduces the delivery rate. Error-corrected distribution instead encodes information to handle specified errors during transmission or storage. Comparing these approaches requires a common target fidelity and a common count of link uses, memories, gates, and classical waiting time.

Detailed study scope. Develop swapping across independently distributed links and compare purification with error-corrected distribution. Calculate the effect of imperfect local operations

and consumed pairs; distinguish heralded success from unconditional rate.

5.3 Loss, heralding, and quantum memories

A link trial succeeds only if all required generation, coupling, transmission, detection, and heralding events occur. Independent efficiencies multiply in a simple model; correlations or multiphoton contamination require a richer description. A geometric waiting-time model is appropriate only when attempts are identically distributed and the trial clock is clearly defined.

A successful pair often waits in memory while another link is created. Its quality then depends on waiting time, not just on its preparation fidelity. A deadline policy discards old pairs, trading lower average age for more repeated attempts. Communication time also limits how quickly a success can be used. These effects couple rate, fidelity, and latency, so multiplying a heralded fidelity by an attempt frequency does not produce a complete network performance metric.

Detailed study scope. Model photon loss, detection efficiency, memory decoherence, and waiting times. Define a link attempt, a clock cycle, and a heralding message. Explain how finite memory lifetime couples link success probability to delivered fidelity.

5.4 Repeaters and routing

A repeater divides a long channel into shorter links and combines locally generated resources. Nested swapping changes waiting-time distributions across levels. Multiplexing increases the number of simultaneous opportunities, while routing determines which stored resources serve a requested connection. Each strategy consumes memory capacity and local-control bandwidth as well as optical attempts.

Scheduling should specify whether successful links wait, are purified, or are discarded, and how contention between applications is resolved. A shortest physical path need not maximize usable entanglement rate if its memories or interfaces are poor.

Compare policies through delivered pairs above a specified fidelity threshold, latency distributions, and resource utilization. An average rate can hide long waiting tails that make a protocol unsuitable for time-sensitive distributed tasks.

Detailed study scope. Compare repeater strategies, nesting, scheduling, multiplexing, and routing under resource limits. Analyze the tradeoff between better pairs and more pairs, and retain local-processing and classical-signalling time in an end-to-end budget.

5.5 Network services and trust

An entanglement service promises a pair to identified endpoints with a fidelity target, delivery signal, and failure behavior. The application then chooses how to use it. Keeping this interface explicit allows a protocol to be analyzed without committing to a particular optical encoding, while still exposing requirements that the physical layer must satisfy.

Trust is part of the service definition. A trusted-node key relay is not equivalent to end-to-end entanglement distributed through untrusted nodes. An untrusted measurement station can still be useful when the protocol and proof account for its behavior. Authenticated classical coordination identifies messages and prevents substitution; it does not eliminate loss or guarantee the underlying quantum state. Security and delivery guarantees must therefore be stated separately and then composed where their assumptions permit.

Detailed study scope. Specify services for pair generation, distributed measurement, and remote computation. Separate physical links, entanglement services, and application demands; identify trusted nodes, untrusted measurement stations, and authenticated classical coordination.

5.6 Distributed quantum tasks

Distributed computation partitions quantum registers across nodes. Local operations are inexpensive relative to remote interactions only under a specified architecture, so the natural

cost units include nonlocal gates, shared pairs, classical rounds, and memory time. A remote gate can consume entanglement and feedforward even when its abstract circuit symbol looks like an ordinary two-qubit gate.

Distributed sensing encodes parameters at separate locations and combines outcomes or shared probes. Its advantage depends on which parameter combination is sought and which local operations are available. Delegation adds a verification or privacy requirement. These tasks should be developed from an explicit input-output specification, followed by a communication protocol and a correctness argument. Entanglement is an enabling resource, not a substitute for the operations and messages that actually perform the task.

Detailed study scope. Develop distributed sensing, delegated operations, and small distributed computations through explicit communication requirements. Explain why sharing entanglement alone does not perform an arbitrary nonlocal gate without further operations and messages.

5.7 Network limits and architecture [advanced]

A network cut separates the desired endpoints into two groups. Any end-to-end communication must cross that cut, so appropriate link-capacity bounds constrain the achievable network rate. Turning such a bound into an achievable protocol additionally requires compatible resources, routing, scheduling, and assistance assumptions; an upper bound is not a deployment recipe.

Memory-assisted strategies can outperform direct transmission in suitable regimes, but finite coherence and local errors can remove the gain. Advanced study compares chain, star, and mesh architectures through common application demands and finite-size simulation. Network coding questions ask whether joint processing changes performance beyond simple routing. Their relevance depends on the allowed quantum and classical operations, and should not be inferred from classical network analogies alone.

Detailed study scope. Compare cut-based rate bounds, memory-assisted strategies, network coding questions, and

finite-size scheduling. Connect network requirements to the optical and memory interfaces developed in Chapter 11, while keeping the protocol analysis hardware independent.

Example 5.1 · Waiting for two elementary links

Let T_1, T_2 be independent geometric waiting times on $1, 2, \dots$ with success probability p per round. With perfect storage, $E[\max(T_1, T_2)] = 2/p - 1/[p(2-p)]$. For $p = 1/2$, the mean is $8/3$ rounds. This excludes swapping failure, memory decay, and communication overhead; these must be added to a usable network model.

Common pitfall

Heralded fidelity is conditional on success. It must be accompanied by delivery probability, rate, latency, and memory assumptions.

5.8 Exercises

Exercise 5.1

Analytical. Use the waiting-time formula at $p = 0.1$ and compare with $1/p$.

Exercise 5.2

Conceptual. Explain the classical messages required by teleportation and swapping.

Exercise 5.3

Diagnostic. Diagnose a repeater-rate estimate that assumes failed memories are available immediately with unchanged fidelity.

Exercise 5.4

Computational. Simulate the two-link waiting process, then add a finite memory deadline and record discarded pairs.

The analytical exercise is discussed in Appendix A.

5.9 Laboratory: A two-link repeater

Build a discrete-event model with independent link attempts, heralding delays, memory decay, and probabilistic swapping. First reproduce the perfect-memory mean in the example within Monte Carlo uncertainty. Then sweep memory lifetime and a discard deadline. Deliver rate-fidelity-latency curves, a complete event definition, and a resource account per delivered pair; no network hardware is required.

Use the submission and validation contract in Appendix C.

References and further reading

- Nielsen, M. A., & Chuang, I. L. (2010). *Quantum computation and quantum information* (10th anniversary ed.). Cambridge University Press.
- Wehner, S., Elkouss, D., & Hanson, R. (2018). Quantum internet: A vision for the road ahead. *Science*, 362(6412), Article eaam9288. <https://doi.org/10.1126/science.aam9288>

6 Quantum cryptography and security

Learning objectives

- Specify security through an adversary model and an operational criterion.
- Trace key distribution from quantum exchange to an authenticated key.
- Distinguish asymptotic illustrations from finite-key guarantees.

Prior QIS chapters: 1, 3, 4, 5.

Foundations reference

Conditional probability, statistical sampling, and measurement. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 3.2, 3.3, 3.5, 6.1. These prerequisite topics are not redeveloped here.

6.1 Security definitions and threat models

A key is correct if the legitimate parties agree, and secret if it is close to uniform and independent of the adversary. A composable secrecy criterion compares the actual joint key-adversary state with an ideal uniform key using trace distance, weighted appropriately when the protocol may abort. Correctness and secrecy failures contribute to a total security budget.

This definition is operational: replacing an ideal key by the real key changes the success probability of any later distinguisher by at most the stated bound. It does not claim that every device automatically meets the mathematical model. Source behavior, detector access, laboratory leakage, random-setting generation, and authentication must be specified. A security proof establishes a conditional guarantee; implementation evidence supports the conditions under which that guarantee is invoked.

Detailed study scope. Introduce correctness, secrecy, trace-distance criteria, and composability. Specify adversarial access, trusted components, authentication assumptions, and failure budgets; distinguish an implementation model from a proof that the model holds.

6.2 Prepare-and-measure and entanglement-based protocols

BB84 encodes random bits in two incompatible bases. After quantum transmission, the parties announce bases, keep compatible events, test a sample, and decide whether to continue. The complementary basis constrains what an adversary could have learned without creating detectable disturbance. An entanglement-based representation helps connect the protocol to correlations and quantum uncertainty.

Intercept-resend illustrates the mechanism but covers only one attack. General security reasoning must allow coherent adversarial interactions, quantum storage, and joint processing within the stated model. The sifted-key error rate summarizes one observable feature of the data; it is not a complete description of adversarial information. The protocol therefore needs a theorem relating the measured statistics and source assumptions to a bound on the adversary's uncertainty.

Detailed study scope. Develop BB84 and an entanglement-based perspective, including sifting, complementary-basis tests, and intercept-resend as an illustrative attack. Explain the relation to entanglement and uncertainty without treating one attack as the general security proof.

6.3 Parameter estimation and reconciliation

Parameter estimation uses disclosed observations to bound properties of the undisclosed raw key. Because the test sample is often selected from a finite record, sampling without replacement and finite-population bounds can be relevant. The choice of confidence parameter controls the chance that the inferred bound is invalid and consumes part of the security budget.

Information reconciliation corrects discrepancies by exchanging public data, such as parity information. Those messages also reveal information to the adversary and must enter the leakage account. Error verification checks whether reconciliation succeeded, usually through an additional public comparison with a bounded collision probability. Authentication protects the transcript from substitution. The sequence of disclosures matters: a key-length calculation must use a model consistent with all information actually made public.

Detailed study scope. Analyze sampling without replacement where appropriate, error estimation, information reconciliation, leakage accounting, and error verification. State which announced data are public and how authentication consumes or refreshes secret resources.

6.4 Privacy amplification and finite keys

Privacy amplification compresses a partially secret raw key into a shorter key using a suitable hash family. A schematic leftover-hashing relation is $\ell \lesssim H_{\min}^{\epsilon}(X | E) - \text{leak}_{\text{EC}} - 2 \log_2(1/\epsilon_{\text{PA}})$, with theorem-dependent constants and security-parameter allocation. The smooth conditional min-entropy quantifies secrecy before the public reconciliation leakage is subtracted under the chosen formulation.

This schematic relation must not be used as a complete protocol proof without its hypotheses and exact terms. Finite samples reduce the entropy estimate, and authentication and verification have their own costs. When the resulting allowable length is nonpositive, the correct action is to abort rather than report the asymptotic rate. A practical teaching calculation should expose each term and explain which theorem would justify it.

Detailed study scope. Introduce universal hashing, smooth entropies, and finite-key corrections. Build a key-length budget with secrecy and correctness parameters. Explain why a positive asymptotic rate need not produce a positive key in a short experiment.

6.5 Optical implementations and side channels

Weak coherent optical pulses can contain multiple photons, so an ideal single-photon model can miss relevant attacks. Decoy-state methods vary source intensity to constrain yields and errors associated with photon-number components under stated source assumptions. They require an estimation model for the observed intensity classes, not merely lower average optical power.

Measurement-device-independent protocols shift detector trust requirements by treating the measurement station as untrusted, while retaining source assumptions. Continuous-variable protocols encode and estimate quadrature information, with security depending on modulation, detection, channel estimation, and reconciliation. Side channels include timing, spectral distinguishability, back-reflections, and control leakage. A protocol family can tolerate some device imperfections while remaining vulnerable to others; the trusted boundary must be stated component by component.

Detailed study scope. Compare single-photon assumptions, weak coherent sources, decoy-state reasoning, detection models, and source leakage. Introduce measurement-device-independent and continuous-variable protocols, with optical tools supplied in Chapter 11 for advanced implementation study.

6.6 Device-independent security and randomness [advanced]

Bell-test data can constrain an adversary without requiring a full calibrated model of the measurement devices. The inference still needs assumptions such as adequate isolation during

a trial, independent setting choices, and a treatment of memory across trials. A finite observed violation must be converted into a quantitative randomness or key bound with the chosen statistical method.

Randomness expansion starts with a seed for settings and processing, then attempts to produce more private randomness than it consumes. Net expansion depends on seed accounting, test rounds, abort probability, and finite-data penalties. A sequence passing ordinary statistical tests is not thereby private from an adversary. Device independence describes which internal models need not be trusted; it does not remove every causal, physical, or statistical assumption.

Detailed study scope. Connect Bell evidence to randomness and device-independent key generation under explicit isolation, independence, and finite-data assumptions. Distinguish random-looking output from certified private randomness and account for seed consumption.

6.7 Cryptographic scope and system integration

Quantum authentication protects quantum data against undetected modification, while quantum secret sharing distributes access rights across participants. Delegated verification asks a client to validate a computation performed elsewhere. Each task has a different ideal functionality, adversary model, and resource budget, so a key-distribution result does not automatically solve it.

Some desired primitives face impossibility results under standard unrestricted models; unconditional bit commitment is a central example. Changing assumptions, such as relativistic separation or restricted storage, creates a different model rather than contradicting the limitation. Post-quantum cryptography uses classical communication with security based on computational assumptions believed to resist quantum attack. QKD uses a physical key-generation protocol with its own implementation and authentication requirements. System design may combine them, but their guarantees should remain explicit.

Detailed study scope. Discuss quantum authentication, secret sharing, delegated verification, and limitations on tasks such as unconditional bit commitment in standard models. Distinguish QKD from post-quantum classical cryptography and require authenticated classical infrastructure in either system design.

Example 6.1 · The disturbance of intercept-resend

In ideal BB84 with uniformly random bases, an interceptor chooses the wrong basis with probability $1/2$. In those cases the sifted bit is wrong with probability $1/2$, giving a sifted-key error rate of $1/4$. This is a diagnostic calculation for one attack, not a finite-key security threshold or a proof against arbitrary attacks.

Common pitfall

A low observed error rate does not alone establish secrecy. Source assumptions, leakage, authentication, finite samples, and the adversary model also enter the claim.

6.8 Exercises

Exercise 6.1

Analytical. If intercept-resend is applied independently to a fraction $f = 0.2$ of ideal transmissions, find its added sifted-key error rate.

Exercise 6.2

Conceptual. Explain why reconciliation leakage must be included before privacy amplification.

Exercise 6.3

Diagnostic. Diagnose a protocol that authenticates public messages using the unauthenticated key from the same exchange.

Exercise 6.4

Computational. Build an illustrative key-budget calculator that separately records sampling, reconciliation leakage, and declared security penalties.

The analytical exercise is discussed in Appendix A.

6.9 Laboratory: From detection records to a key budget

Simulate ideal BB84 records with a controllable intercept-resend fraction and sample a test subset. Validate the expected sifting fraction and $f/4$ disturbance with sampling intervals. Produce a transparent illustrative key budget using explicitly supplied finite-key terms; mark it as a teaching model unless a complete security theorem and its hypotheses are implemented. Deliver the public transcript categories and leakage account. Use the submission and validation contract in Appendix C.

References and further reading

- Portmann, C., & Renner, R. (2022). Security in quantum cryptography. *Reviews of Modern Physics*, 94(2), Article 025008. <https://doi.org/10.1103/RevModPhys.94.025008>
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PART III

Computation and scientific applications

7 Computation models, compilation, and complexity

Learning objectives

- Relate computation models through their resources and restrictions.
- Compile reversible computations with ancilla and connectivity costs.
- Interpret speedup and hardness claims with explicit access models.

Prior QIS chapters: 1.

Foundations reference

Elementary gates, circuit order, unitary evolution, and basic annealing. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 8.1, 8.2, 8.3, 10.1, 12.3. These prerequisite topics are not re-developed here.

7.1 Reversible computation and coherent access

A classical function can be embedded reversibly as $|x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle$. A real implementation may create temporary work registers. If they remain correlated with x , discarding them can destroy the coherence needed by later interference. Computing,

using the result coherently, and uncomputing the workspace restores clean ancillas.

Phase kickback uses an eigenstate of the target action to place a phase on the control register. It turns a function evaluation into a phase oracle under specific preparation and access assumptions. This is an algorithmic construction beyond the elementary gate definitions in Foundations. Resource analysis must count reversible arithmetic, workspace, data access, and cleanup, rather than treating the oracle symbol as a free operation with unlimited input size.

Detailed study scope. Develop reversible embeddings of classical functions, clean ancillas, uncomputation, and phase kickback as computational mechanisms. Distinguish an oracle specification from the cost of constructing and calling that oracle.

7.2 Universality and gate synthesis

A universal gate set can approximate arbitrary finite-register unitaries to a chosen tolerance. With a finite alphabet, exact representation and approximation are different questions. The precision needed by a complete algorithm determines synthesis cost, and approximating many operations requires allocating an error budget across the circuit.

A synthesis method decomposes a target into allowed operations while controlling error in a specified norm. Clifford operations plus a non-Clifford resource form a common logical setting, but physical native gates can differ. Compilation therefore has at least two levels: expressing the algorithm in a logical library and implementing that library on an architecture. Gate counts should report which level they describe, how errors compose, and whether state preparation and measurement are included.

Detailed study scope. Introduce universal gate sets, approximate synthesis, precision dependence, and logical gate counts. Use the elementary gates already known from Foundations without listing their matrices again. Separate exact universality from approximation over a finite gate alphabet.

7.3 Alternative computation models

Measurement-based computation prepares an entangled graph resource and performs local measurements whose choices can depend on earlier outcomes. Byproduct operators are tracked or corrected through classical feedforward. The computation lies in the resource and adaptive measurement pattern, so replacing every adaptive choice by a fixed basis can change the task.

Adiabatic computation encodes a problem into an evolving Hamiltonian and relies on a sufficiently controlled path and spectral structure. Its relation to circuit computation concerns formal models with specified resources and precision, not a guarantee that an arbitrary annealer efficiently solves a hard optimization problem. The basic annealing interpolation belongs to Foundations. Here the new questions are model equivalence, locality, gap dependence, simulation overhead, and which restrictions alter computational power.

Detailed study scope. Develop measurement-based computation on graph states and compare it with circuit and adiabatic computation. State the assumptions of model equivalence; refer to Foundations for the introductory annealing interpolation instead of deriving it again.

7.4 Compilation and routing

Routing maps logical interactions onto a physical connectivity graph. A SWAP changes where information resides, so the logical-to-physical map must be updated after each routing operation. A final measurement string can otherwise be interpreted in the wrong register order even when the compiled circuit is physically correct.

Scheduling then assigns compatible operations to time steps under gate duration, crosstalk, measurement, and feedforward constraints. Minimizing gate count need not minimize depth or error: a longer parallel circuit can finish sooner than a

shorter serial one. A useful compiler comparison fixes the native gate set, connectivity, approximation tolerance, and objective. Equivalence checks should test coherent action, including ancillas and output permutations, rather than only a few computational-basis inputs.

Detailed study scope. Analyze decomposition, connectivity-aware placement, routing, scheduling, and measurement feed-forward. Track logical-to-physical permutations and compare depth, two-qubit count, ancilla use, and communication latency under a declared native instruction set.

7.5 Complexity classes and reductions

P describes polynomial-time deterministic classical decision procedures; BPP allows bounded-error randomness, and BQP allows bounded-error quantum computation. NP concerns efficiently checkable classical witnesses, while QMA permits a quantum witness verified by an efficient quantum process with completeness and soundness conditions. Promise problems restrict which inputs require a specified answer.

Known containments must be separated from conjectured separations. In particular, lack of a known efficient algorithm does not prove that an NP-complete problem is outside BQP. Reductions transfer difficulty by converting one problem into another with bounded overhead. A runtime measured on a few instances is evidence about those instances and implementations, not a proof of a complexity-class separation or an average-case hardness claim.

Detailed study scope. Introduce P, BPP, BQP, NP, and QMA at the level needed to interpret algorithms and verification. Distinguish decision and promise problems, reductions, worst-case complexity, and average-case performance. State which separations are proved and which are conjectured.

7.6 Classical simulation and verification

Strong simulation computes output probabilities or amplitudes to a specified precision; weak simulation samples from a distribution. These tasks can have different costs. Clifford circuits with stabilizer inputs admit efficient classical simulation, while tensor networks can exploit bounded entanglement or favorable graph structure. Large qubit count alone therefore does not imply classical intractability.

Verification asks whether an output or distribution meets a claim. For some problems, checking a proposed solution is much cheaper than generating it; for sampling tasks, reliable verification can itself be difficult. A benchmark should state what its test detects and what it leaves untested. Comparing a quantum circuit with a naive full-state simulator is insufficient when a structure-exploiting classical algorithm applies to the same instances.

Detailed study scope. Develop stabilizer, tensor-network, and low-entanglement simulation regimes conceptually, with stabilizer mechanics deferred to Chapter 12. Compare strong and weak simulation, output verification, and sampling tasks with computationally costly probability evaluation.

7.7 Resource accounting and advantage claims [advanced]

An end-to-end cost includes constructing the input, accessing data, building oracles, executing coherent steps, amplifying success, measuring the requested output, and classical postprocessing. Query complexity isolates one part of this ledger to reveal algorithmic structure; it should be reported as such.

Precision can dominate costs in scientific tasks, and output size can dominate costs when a full classical vector is requested. A proposed advantage must use comparable error criteria and access assumptions on both sides. Empirical comparisons additionally require reproducible hardware or simulator conditions

and a competent classical baseline. Formal speedup results remain valuable even before practical benefit is established, provided the model and its limitations are visible.

Detailed study scope. Combine input loading, oracle construction, precision, success amplification, output extraction, and classical preprocessing into an end-to-end cost. Separate query advantage, asymptotic runtime advantage, and measured application benefit.

Example 7.1 • Connectivity changes implementation cost

A compiled logical circuit contains twelve CNOT gates. Routing inserts four SWAP gates. If each SWAP uses three native CNOTs, the resulting two-qubit count is $12 + 4 \times 3 = 24$. This count assumes that no subsequent cancellation occurs; depth and error still depend on scheduling and the device model.

Common pitfall

An oracle call is an access-model unit, not automatically a constant-time physical operation. Query complexity alone does not establish practical speedup.

7.8 Exercises

Exercise 7.1

Analytical. A circuit has twenty CNOTs and routing adds seven three-CNOT SWAPs. Find the new count and routing fraction.

Exercise 7.2

Conceptual. Explain why leaving an ancilla entangled can invalidate a reversible subroutine.

Exercise 7.3

Diagnostic. Assess a claim that an NP-complete problem lies outside BQP because no efficient quantum algorithm is known.

Exercise 7.4

Computational. Compare two placements for a small interaction graph and report two-qubit count and depth separately.

The analytical exercise is discussed in Appendix A.

7.9 Laboratory: Compilation with explicit costs

Implement a reversible three- or four-bit predicate with clean ancillas, then compile it for all-to-all and linear connectivity. Verify the full coherent action on superpositions, including ancilla reset, up to a global phase. Deliver a resource table before and after routing, the output permutation, and a comparison with direct classical evaluation. Fix the synthesis tolerance and count any preparation overhead.

Use the submission and validation contract in Appendix C.

References and further reading

- Nielsen, M. A., & Chuang, I. L. (2010). *Quantum computation and quantum information* (10th anniversary ed.). Cambridge University Press.
- Preskill, J. (n.d.). *Lecture notes for Physics 219/Computer Science 219: Quantum computation*. California Institute of Technology. Retrieved September 5, 2026, from <https://www.preskill.caltech.edu/ph229/>
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8 Quantum algorithms

Learning objectives

- Derive the mechanisms behind the principal algorithm families.
- State oracle promises, precision, and success probability.
- Assess algorithms through complete resource and output requirements.

Prior QIS chapters: 7.

Foundations reference

Relative phase and circuit composition. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 2.5, 8.3. These prerequisite topics are not redeveloped here.

8.1 Interference and oracle algorithms

Oracle algorithms expose how interference extracts a global property of a promised function. Deutsch–Jozsa distinguishes constant from balanced functions in an exact query setting; Bernstein–Vazirani retrieves a hidden linear bit string; Simon yields linear constraints on a hidden XOR period. Their promises, outputs, and comparison models differ, so their speedups should not be summarized by one undifferentiated claim.

The shared mechanism is coherent evaluation followed by interference that suppresses outputs incompatible with the hidden structure. Simon's procedure, for example, produces random equations orthogonal to the hidden period, which classical linear algebra then solves. This separates the quantum information-gathering step from classical reconstruction. The oracle must implement the promised function coherently; inspecting a simulator's stored truth table is not an allowed shortcut in the query model.

Detailed study scope. Use Deutsch–Jozsa, Bernstein–Vazirani, and Simon as a connected progression in promise problems, phase kickback, and hidden structure. Compare exact and bounded-error query models and identify where the promise enters each argument.

8.2 Quantum Fourier transform

For $N = 2^n$, the quantum Fourier transform maps $|x\rangle$ to $N^{-1/2} \sum_{y=0}^{N-1} e^{2\pi i xy/N} |y\rangle$. Its circuit factorizes the phase structure into controlled rotations and basis changes. The usual decomposition also reverses output order unless swaps or classical relabeling correct it.

Dropping small controlled rotations creates an approximate transform with lower implementation cost. The omitted angles must be related to an overall error tolerance and to the surrounding algorithm's success criterion. The transform acts coherently on amplitudes, but a measurement returns one sample from the resulting distribution. Its algorithmic value comes from concentrating useful information into accessible outcomes, as in period finding, rather than delivering an entire classical Fourier table at once.

Detailed study scope. Develop the discrete quantum Fourier transform, controlled-phase decomposition, inverse transform, output order, and approximate truncation. Relate its action to periodic structure without identifying it with classical access to every Fourier coefficient.

8.3 Phase estimation

For an eigenstate $U|u\rangle = e^{2\pi i\phi}|u\rangle$, controlled powers imprint phases on a control register, and the inverse Fourier transform converts them into an estimate of ϕ . With m bits, an exactly representable phase gives a definite ideal output. Otherwise the output is a peaked distribution whose tails determine the probability of a useful approximation.

For a superposition of eigenstates, the result samples eigen-phases according to the input overlaps and leaves a corresponding conditional target state. State preparation can therefore be a bottleneck. Powers U^{2^j} are costly unless a special implementation provides them efficiently; counting them as one generic gate hides that cost. Energy estimation also requires translating phase precision through the evolution time and resolving phase aliasing.

Detailed study scope. Develop coherent and iterative phase estimation, finite-bit resolution, eigenstate overlap, and success amplification. Count controlled powers and their implementation cost. Explain phase wraparound and the distinction between energy precision and binary register size.

8.4 Search and amplitude amplification

Grover amplification acts in the span of the normalized good and bad components. If the initial good probability is $a = \sin^2 \theta$, each iteration rotates by 2θ , giving success $\sin^2((2r+1)\theta)$ after r iterations. This derives both the quadratic query scaling and the loss from continuing past the optimal rotation.

General amplitude amplification replaces uniform preparation with an arbitrary efficiently reversible preparation and its inverse. If the number of marked items is unknown, iteration selection or estimation must handle uncertainty. The quadratic advantage is an oracle-model result; constructing a membership test, preparing the search space, and verifying outputs still cost resources. A structured classical search method can outperform an unstructured baseline when the input offers exploitable information.

Detailed study scope. Derive the two-dimensional rotation picture of Grover search and general amplitude amplification. Address multiple or unknown marked items, overshooting, oracle construction, and the quadratic query lower-bound setting.

8.5 Amplitude estimation and Monte Carlo tasks

Amplitude estimation estimates the success probability of a coherently prepared procedure. In the standard coherent query setting it can achieve additive precision with query scaling of order $1/\epsilon$, compared with order $1/\epsilon^2$ for independent direct sampling at fixed confidence. Constants, confidence amplification, and implementation assumptions remain part of the comparison.

The gain requires coherent access to preparation, its inverse, and suitable reflections or equivalent controlled procedures. Iterative variants trade control-register structure and phase-estimation machinery for a sequence of experiments and classical inference. Noise limits usable depth, and difficult data preparation can dominate the saved samples. A Monte Carlo application should therefore compare total cost for the same observable, confidence, bias, and input-access model, including classical preprocessing and output verification.

Detailed study scope. Connect amplitude estimation to probability estimation, precision scaling, and repeated coherent access. Compare coherent and iterative variants through depth, shots, confidence, and state-preparation costs; avoid treating sampling complexity as total runtime.

8.6 Period finding and Shor's algorithms

Shor's factoring procedure reduces factoring to finding the order r of a chosen integer a modulo N . Quantum period finding estimates fractions related to s/r through modular exponentiation and Fourier structure. Continued fractions and classical checks recover candidate orders; suitable even orders then produce factors through greatest-common-divisor calculations.

Not every random choice of a or every measurement succeeds, so the full algorithm includes retries and classical verification. Reversible modular arithmetic usually dominates resource estimates more than the final Fourier transform. Discrete-logarithm algorithms exploit related hidden-group structure but require their own arithmetic and group representation. The correct accounting includes register sizes, arithmetic ancillas, synthesis precision, success probability, and, for a physical implementation, fault-tolerant overhead.

Detailed study scope. Develop order finding, modular arithmetic, continued-fraction recovery, and the classical steps needed for factoring. Explain discrete logarithms at the algorithm-family level and separate arithmetic circuit resources from Fourier-transform cost.

8.7 Quantum walks and algorithm design [advanced]

A continuous quantum walk uses a graph-related Hamiltonian, while a discrete walk uses a unitary step often combining a coin or reflection structure with movement. Spectral properties determine propagation, mixing-like behavior, and search performance. Quantum evolution is unitary, so it does not simply reproduce the convergence of a classical stochastic walk.

Algorithm design connects a problem to a graph or operator whose spectrum reveals the desired property. This can lead to search, element-distinctness-type tasks, or matrix-function constructions, but each improvement depends on an access model. The transition to Chapter 9 generalizes this viewpoint: instead of treating a circuit as a list of gates, one engineers a polynomial or spectral transformation and then counts the cost of realizing its access primitives and extracting its answer.

Detailed study scope. Introduce discrete and continuous quantum walks, spectral structure, and applications to search and graph tasks. Connect these ideas to the matrix-function techniques in Chapter 9. Build an algorithm-selection map based on input access, requested output, and verifiable performance.

Example 8.1 · An exactly representable phase

If $U|u\rangle = e^{2\pi i(3/8)}|u\rangle$, ideal three-bit phase estimation with input $|u\rangle$ returns 011 with certainty under the stated most-significant-bit convention. The controlled powers U, U^2, U^4 require seven uses of U when implemented by repetition. An exact binary phase removes discretization error, not implementation cost.

Common pitfall

A Fourier-transform circuit does not expose its complete amplitude array. Reading many output coefficients can remove the claimed computational saving.

8.8 Exercises

Exercise 8.1

Analytical. For phase $5/8$, state the ideal three-bit output and the repeated- U cost. What changes with four phase bits?

Exercise 8.2

Conceptual. Explain why Grover search can lose success probability after too many iterations.

Exercise 8.3

Diagnostic. Diagnose a factoring estimate that counts QFT gates but omits modular exponentiation.

Exercise 8.4

Computational. Compare phase-estimation distributions for an exact phase and a nearby nonrepresentable phase, with controlled-power costs included.

The analytical exercise is discussed in Appendix A.

8.9 Laboratory: Phase estimation and amplification

Implement ideal phase estimation and Grover amplification on at most eight qubits. Validate exact-phase outputs and the success law $\sin^2((2r+1)\theta)$ for known marked fraction $\sin^2\theta$. Sweep precision bits and Grover iterations; then add a declared noise channel. Deliver success-versus-cost curves, including oracle construction assumptions and classical comparison costs. Use the submission and validation contract in Appendix C.

References and further reading

- Childs, A. M. (n.d.). *Lecture notes on quantum algorithms*. University of Maryland. Retrieved September 5, 2026, from <https://www.cs.umd.edu/~amchilds/qa/>
- Montanaro, A. (2016). Quantum algorithms: An overview. *npj Quantum Information*, 2, Article 15023. <https://doi.org/10.1038/npjqi.2015.23>
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9 Quantum simulation and scientific computing

Learning objectives

- Select simulation and matrix-function methods for a specified access model.
- Map physical problems to computational representations.
- Evaluate approximation, preparation, measurement, and classical comparison costs.

Prior QIS chapters: 1, 7, 8.

Foundations reference

Spectral methods, Hamiltonian evolution, oscillator states, and the elementary Ising model. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 1.7, 10.1, 10.2, 11.3, 12.1. These prerequisite topics are not redeveloped here.

9.1 Digital Hamiltonian simulation

For $H = A + B$, a first-order product formula approximates e^{-itH} by $(e^{-itA/r} e^{-itB/r})^r$. Noncommutativity produces a leading error governed by commutators, scaling as t^2/r under fixed bounded-operator assumptions. A symmetric second-order step cancels the leading asymmetry and gives a higher-order small-step approximation.

This is a new algorithmic question beyond evaluating a small matrix exponential. The number of factors, their native implementation, Hamiltonian decomposition, and ordering affect cost. An operator-norm error bounds every input, while an observable error may be much smaller for one state and time. Both are useful but answer different questions. Convergence studies should fix total simulated time and identify the regime in which the expected asymptotic scaling is actually observed.

Detailed study scope. Develop product formulas for noncommuting terms, commutator-dependent error, higher-order constructions, and ordering effects. Compare them with exact small-system propagation only as a validation reference; elementary matrix-exponential evolution remains in Foundations.

9.2 Block encodings and linear combinations of unitaries

A block encoding places a scaled matrix A/α in a selected ancilla block of a unitary: $(\langle 0| \otimes I)U(|0\rangle \otimes I) \approx A/\alpha$. The normalization α , ancilla size, approximation tolerance, and cost of U are part of the representation. A favorable polynomial transformation is not useful if obtaining this access is prohibitively expensive.

Linear combinations of unitaries construct such blocks by preparing a coefficient superposition, selecting a controlled unitary, and unpreparing the ancilla. Postselection or amplitude amplification handles the desired block, introducing success and normalization costs. Sparse-entry or state-preparation oracles are alternative access models, each requiring a realistic implementation account. The matrix dimension alone does not determine the speedup.

Detailed study scope. Define a block encoding with its normalization, ancillas, and approximation error. Develop linear-combination-of-unitaries constructions and explain how access to matrix entries or state-preparation circuits determines the actual cost.

9.3 Qubitization and singular value transformation [advanced]

Qubitization organizes a block-encoded operator into invariant subspaces where controlled reflections produce spectral rotations. Quantum singular value transformation then interleaves phases with block-encoding access to implement a constrained polynomial of singular values. Polynomial parity and boundedness conditions are mathematical requirements for the resulting sequence to remain unitary.

Applications reduce to approximation problems: approximate a sign, inverse, exponential-related function, or spectral filter on a specified interval. The polynomial degree controls query cost, while a spectral gap or condition number determines how hard the approximation is near problematic points. Full constructive proofs are an advanced extension; the core operational chain is target function, allowed spectral interval, polynomial error, access cost, and observable readout. This chain prevents an abstract matrix transformation from being mistaken for free classical matrix arithmetic.

Detailed study scope. Introduce invariant subspaces, polynomial transformations, parity and boundedness constraints, and quantum singular value transformation. Connect polynomial degree to precision, spectral gaps, and condition number rather than presenting exponential improvement without an access model.

9.4 Quantum linear systems and spectral tasks

A quantum linear-system algorithm aims to prepare a state proportional to $A^{-1}|b\rangle$, typically for an invertible Hermitian matrix or a suitable embedding. Small eigenvalues are amplified by inversion, so condition number and normalization affect both precision and success cost. Preconditioning can improve the spectrum but itself needs an efficient accessible representation.

The useful output is often an expectation value or overlap of the solution state. Recovering every classical component generally requires many measurements and can erase the advantage. A fair comparison therefore fixes the requested statistic and the input-access model rather than comparing state preparation with printing a full classical vector. Singular systems, unknown normalization, and approximate data require additional problem specifications; they cannot be hidden inside the inverse notation.

Detailed study scope. Develop state preparation proportional to a matrix inverse applied to an input, observable estimation, and condition-number dependence. Distinguish preparing a solution state from outputting every classical solution component and compare with preconditioned classical methods.

9.5 Fermions and electronic structure

Second quantization describes fermions through occupation numbers and creation and annihilation operators obeying anticommutation relations. Electronic Hamiltonians contain one- and two-body terms built from these operators. Mapping occupations to qubits requires preserving fermionic signs: Jordan–Wigner uses parity strings, while Bravyi–Kitaev redistributes occupation and parity information.

A chemistry calculation also chooses a basis, active space, electron sector, and target accuracy. Symmetries can reduce resources, but omitted orbitals and approximate integrals introduce scientific modeling error before any quantum algorithm runs. State preparation, phase estimation or variational optimization, and Hamiltonian measurement then add algorithmic costs. Comparing methods requires the same molecular model and error target, with tractable instances validated against established classical electronic-structure calculations.

Detailed study scope. Introduce second quantization, occupation encoding, Jordan–Wigner and Bravyi–Kitaev mappings, symmetries, active spaces, and molecular observables. Account for integral generation, state preparation, measurement, and precision in a chemistry resource estimate.

9.6 Many-body and analog simulation

Many-body simulation targets correlations, phases, transport, or nonequilibrium evolution rather than an entire inaccessible wavefunction. Entanglement growth can make some tensor-network simulations expensive, but geometry, time, observables, and state structure determine the actual classical difficulty. A noncommuting interacting model provides a meaningful extension beyond the elementary Ising spectrum in Foundations.

An analog simulator realizes a controllable physical Hamiltonian directly. Its limitations include unwanted interactions, calibration uncertainty, finite temperature, and imperfect preparation and readout. Validation can combine small-instance exact results, symmetry checks, conserved quantities, and comparisons across methods. Agreement in one observable does not establish every property of the implemented dynamics. A scientific claim should identify which quantities are robust to the known model discrepancies.

Detailed study scope. Develop analog simulator validation, lattice models beyond the introductory Ising example, entanglement growth, nonequilibrium dynamics, and tensor-network comparisons. Separate a physical analog model from a universal digital algorithm and state the control restrictions.

9.7 Error budgets and scientific evidence

Choose a target quantity before assembling an error budget. For an energy estimate, errors may arise from basis truncation, state preparation, approximate evolution, imperfect hardware, and finite measurement. Some are bounded biases and others statistical uncertainties; they should not be added indiscriminately as if they were identical probabilities.

A defensible workflow allocates tolerances, demonstrates convergence of numerical approximations, propagates uncertainty, and compares with an independent small-instance reference. Increasing circuit depth can reduce algorithmic approximation error while increasing physical noise. Increasing shots reduces sampling error but cannot remove systematic model bias. The

scientifically useful operating point balances these contributions under a fixed resource budget and states which residual errors remain uncontrolled.

Detailed study scope. Partition state-preparation, discretization, algorithmic, sampling, and hardware errors in a compatible target quantity. Compare energy, correlation, and dynamical-observable goals; require convergence studies and independent classical checks for tractable instances.

Example 9.1 · A solution state is not a classical vector readout

For $A = \text{diag}(1, 2)$ and $|b\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, the normalized state proportional to $A^{-1}|b\rangle$ is $(2|0\rangle + |1\rangle)/\sqrt{5}$. Computational-basis probabilities are $4/5$ and $1/5$. Producing this state does not report every signed component of a large classical solution vector.

Common pitfall

Efficient state preparation does not imply efficient extraction of an arbitrary classical output. Data access, condition number, precision, and readout belong in the same comparison.

9.8 Exercises

Exercise 9.1

Analytical. For $A = \text{diag}(1, 3)$ and the same input, find the normalized solution state and its first-outcome probability.

Exercise 9.2

Conceptual. Explain how a block-encoding normalization changes algorithmic cost.

Exercise 9.3

Diagnostic. Diagnose a simulation accuracy claim based only on agreement at one time step.

Exercise 9.4

Computational. Compare first- and second-order product formulas for noncommuting terms using operator error and an observable error.

The analytical exercise is discussed in Appendix A.

9.9 Laboratory: Simulation error at fixed resources

Use a three- or four-site noncommuting spin Hamiltonian and compare first- and second-order product formulas against exact propagation at the same total time. Sweep step count, report operator error and a correlation observable, and count exponentials or native gates. Validate the expected small-step convergence regime and explain departures. Optionally map a two-orbital fermionic problem; do not repeat the elementary two-spin spectrum exercise.

Use the submission and validation contract in Appendix C.

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10 Variational methods, optimization, and quantum learning

Learning objectives

- Design variational objectives and measurement budgets.
- Evaluate trainability, optimization, and learning performance.
- Compare hybrid methods with explicit classical baselines and data costs.

Prior QIS chapters: 1, 7, 8, 9.

Foundations reference

Sampling uncertainty, Ising models, QUBO formulation, and introductory annealing. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 3.5, 12.1, 12.2, 12.3. These prerequisite topics are not redeveloped here.

10.1 Hybrid variational workflows

A variational method prepares $\rho(\theta)$, estimates a cost $C(\theta)$, and updates parameters classically. The optimization objective is therefore observed through a measurement process, often with

noise and bias. The cost of a run is the sum over all evaluations, gradients, restarts, calibration, and final validation, not the number of parameters or iterations alone.

An ansatz defines the accessible state family, while the optimizer searches that family. Increasing expressivity can improve representational capacity but worsen trainability. A stopping rule should distinguish optimizer stagnation from statistical uncertainty and insufficient expressivity. Record exact simulator optima only as reference information; the algorithm being assessed must use the same measurements and data access that its stated implementation allows.

Detailed study scope. Specify the ansatz, objective, measurement strategy, optimizer, initialization, and stopping rule. Separate the cost of one expectation estimate from a full optimization run and include repeated starts, validation, and classical preprocessing.

10.2 Variational eigensolvers and excited states

For a normalized trial state, the exact energy expectation is at least the ground-state energy. VQE uses this variational principle to turn state preparation into an optimization problem. Decomposing the Hamiltonian into measurable terms makes the objective accessible, but grouping, covariance, and shot allocation determine its measurement cost.

A noisy finite-sample estimate may fall below the true ground energy, so the upper-bound statement applies to the exact expectation, not every reported sample mean. Symmetry constraints restrict the search to a relevant sector and can prevent unphysical solutions. Excited-state methods add orthogonality penalties, subspace diagonalization, or other constraints, each with extra measurement and conditioning costs. A chemical or physical application must also separate ansatz error from the error in the underlying model Hamiltonian.

Detailed study scope. Develop the variational principle as an algorithmic objective, Hamiltonian measurement grouping, symmetry constraints, and extensions to excited states and dynamics. Distinguish an energy upper bound for an exact expectation from a noisy estimator.

10.3 QAOA and constrained optimization

QAOA alternates a cost-generated unitary with a mixer, producing a depth-parameterized variational family. The cost operator comes from the already established problem encoding; the new design question is how the mixer, initialization, and angles explore the feasible space. A constraint-preserving mixer can avoid invalid states but may introduce more complex interactions.

Approximation ratio, feasibility probability, solution quality, and sampling cost are distinct outputs. Shallow success on one small graph does not imply scalable performance. Compare against classical local search, exact solvers where feasible, and problem-specific heuristics under declared budgets. Basic QUBO-to-Ising conversion and introductory annealing schedules remain references to Foundations, while this section develops coherent alternating-operator design and its optimization landscape.

Detailed study scope. Develop alternating-operator ansatz design, mixer choices, feasibility-preserving encodings, depth, and approximation metrics. Assume the Ising/QUBO mapping from Foundations and focus on the algorithm, landscape, and comparison with classical heuristics.

10.4 Gradients and trainability

When a parameter enters through $e^{-i\theta P/2}$ with $P^2 = I$, an expectation derivative obeys the parameter-shift relation $\partial_\theta f = [f(\theta + \pi/2) - f(\theta - \pi/2)]/2$. The rule follows from the two-frequency dependence generated by the two eigenvalues. Other generators can require different shifts or multiple evaluations.

Independent shot noise in the two shifted estimates adds in variance. Allocating more samples to high-variance terms can improve a fixed-budget gradient. Barren plateaus describe regimes in which relevant gradient variance becomes very small, often with system size; noise can create additional concentration. These are model-dependent results, not a claim that all variational training always fails. Architecture, locality, initialization, and cost structure influence the landscape.

Detailed study scope. Derive parameter-shift rules under their generator assumptions, compare stochastic gradients and finite differences, and analyze shot allocation. Introduce barren plateaus, noise-induced concentration, expressivity, and the tradeoff between ansatz flexibility and trainability.

10.5 Quantum data and feature representations

A quantum learning model may receive quantum states directly or encode classical vectors into states. These settings have different data-access costs. Amplitude encoding can be compact in qubit count while expensive to prepare; an angle embedding can be cheaper yet express a different feature family. Access to a simulator array is not equivalent to experimental access to a quantum state.

A kernel estimates similarity between feature states, often through an overlap or related observable. Estimation introduces sampling error, and the number of pairwise comparisons can be large. The relevant question is whether the representation exposes useful structure at a competitive total cost. A high-dimensional Hilbert space alone does not guarantee useful features, improved generalization, or a computational advantage over an efficiently computed classical kernel.

Detailed study scope. Distinguish quantum-native data from classical data embedded into a circuit. Account for state preparation, feature maps, kernels, accessible observables, and the measurement cost of a similarity estimate.

10.6 Learning models and generalization

Training minimizes empirical loss, while generalization concerns unseen data from the target distribution. Data splitting, preprocessing, and hyperparameter selection must respect this distinction. Tuning a quantum feature map repeatedly on the test set invalidates the test just as it does for a classical model.

Model capacity, sample size, label noise, and concentration of quantum features affect generalization. Quantum-native learning can use collective or adaptive measurements unavailable to a fixed classical dataset, but the comparison must allow the classical method access to the specified measurement record. Report repeated runs, uncertainty, and competent baselines. A favorable result may establish an empirical benefit on a stated dataset without supporting a universal learning advantage or a scaling theorem.

Detailed study scope. Develop variational classifiers, quantum kernels, and learning from quantum experiments. Discuss training versus test performance, model capacity, concentration, data leakage, and sample complexity; require fair preprocessing and hyperparameter budgets for classical baselines.

10.7 Applications and limitations [advanced]

Near-term variational methods are often heuristics: their performance is established by analysis in restricted settings and by experiments on selected instances. Fault-tolerant algorithms can offer different guarantees but introduce state preparation, coherent depth, and error-correction costs. These two regimes should not share an advantage claim without a common resource model.

A useful application study states the input distribution, requested output, precision, hardware assumptions, and classical comparator. It then separates proved scaling, model-based estimates, and measured results. Negative results are informative when they locate the bottleneck: loading data, trainability, shot noise, depth, or readout. This makes the chapter a method for evaluating optimization and learning proposals, rather than a

catalogue of applications asserted to benefit from quantum processing.

Detailed study scope. Compare near-term heuristics with fault-tolerant subroutines in optimization and learning. Assess scaling with data dimension, precision, depth, and noise; distinguish provable restricted-model results from empirical results on small demonstrations.

Example 10.1 · The measurement cost of a parameter-shift gradient

For a suitable two-eigenvalue generator and a $\pi/2$ shift, $g = [f(\theta + \pi/2) - f(\theta - \pi/2)]/2$. Independent estimates 0.6 and 0.2, each with standard error 0.04, give $g = 0.2$ and standard error $\sqrt{0.04^2 + 0.04^2}/2 \simeq 0.0283$. Correlated estimates require a covariance term.

Common pitfall

A successful fit on a small benchmark does not establish quantum advantage. Training budget, data loading, test data, noise, and classical baselines affect the comparison.

10.8 Exercises

Exercise 10.1

Analytical. For shifted estimates 0.3 and -0.1 with independent standard errors 0.02, compute the gradient and its standard error.

Exercise 10.2

Conceptual. Explain why a constraint-preserving mixer can change the accessible search space.

Exercise 10.3

Diagnostic. Diagnose a learning comparison that tunes the quantum model on the test set.

Exercise 10.4

Computational. Compare exact and sampled variational gradients as shot count changes, reporting total objective evaluations.

The analytical exercise is discussed in Appendix A.

10.9 Laboratory: A variational algorithm with a fair budget

Choose either a four-qubit VQE or a small constrained QAOA problem whose mapping is supplied. Compare exact expectations with shot-based optimization across at least five recorded seeds. Validate the exact objective against diagonalization or enumeration and compare with a classical method under a stated evaluation budget. Deliver convergence, final-error distributions, gradient costs, and a held-out noise or instance check.

Use the submission and validation contract in Appendix C.

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PART IV

Reliable physical systems

11 Physical architectures and photonic information

Learning objectives

- Compare physical platforms through common information-processing requirements.
- Use optical and Gaussian descriptions for communication and encoding.
- Translate application demands into control, memory, and interface constraints.

Prior QIS chapters: 1, 2, 5.

Foundations reference

Two-level dynamics, position and momentum, and the quantum harmonic oscillator. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 10.3, 10.4, 11.1, 11.3. These prerequisite topics are not redeveloped here.

11.1 Architecture requirements and comparison

A platform must prepare, manipulate, couple, measure, reset, and preserve information over the time needed by an application. Native connectivity determines routing; measurement

and reset affect syndrome cycles; leakage and crosstalk affect whether a simple independent error model is adequate. These properties interact, so a single best gate number cannot summarize an architecture.

Comparison begins with a common task, such as a repeated error-correction cycle or entanglement delivery between modules. The resulting requirement matrix includes duration, conditional and unconditional success, error channels, feedforward latency, and scaling constraints. Resource estimates should distinguish measured component performance from simultaneous integrated operation. The goal is to understand physical tradeoffs, not to assign a universal ranking to platforms with different strengths and operating conditions.

Detailed study scope. Define native interactions, connectivity, coherence, reset, readout, leakage, control bandwidth, and modularity. Compare a complete operation cycle and its error channels rather than ranking platforms by an isolated qubit count or a best-case gate fidelity.

11.2 Superconducting circuits and semiconductor spins

Superconducting circuits encode information in engineered circuit modes with controlled anharmonicity. Coupling and microwave control enable operations, while relaxation, dephasing, leakage, and readout backaction shape errors. Semiconductor spins use spin degrees of freedom and electrically or magnetically controlled interactions, with material, charge, and nuclear environments affecting coherence.

The information-processing description connects these mechanisms to channels and control constraints rather than repeating two-level dynamics. Cryogenic wiring, amplifier chains, fabrication variation, and calibration overhead limit integrated operation. A tunable interaction can improve programmability while increasing sensitivity to noise. Architecture analysis should follow the complete control and measurement path and identify which assumptions about isolation, reset, and crosstalk are needed by the computational model.

Detailed study scope. Develop the physical origin of computational encodings, tunability, coupling, addressability, and characteristic noise. Connect fabrication variability, control electronics, cryogenic constraints, and readout to system-level resource requirements.

11.3 Trapped ions, neutral atoms, and defects

Trapped-ion systems can mediate interactions through collective motion, linking gate performance to mode control, heating, and addressing. Neutral-atom systems use optical trapping and, in important architectures, Rydberg-mediated interactions; geometry, rearrangement, atom loss, and blockade imperfections affect operations. Defect-based systems can connect long-lived local degrees of freedom to optical interfaces.

These mechanisms create different tradeoffs among connectivity, parallelism, memory, and networking. Transport or rearrangement consumes time and can introduce errors; all-to-all effective connectivity may still face mode or scheduling constraints. A fair comparison specifies the gate or network operation being implemented and includes preparation, measurement, and reset. Historical component demonstrations motivate models but should not be presented as current performance specifications without a dated source.

Detailed study scope. Compare collective motion, Rydberg interactions, optical control, and solid-state defect interfaces. Analyze transport or rearrangement, crosstalk, loss, and memory-network tradeoffs through the operational tools of Chapters 1–2.

11.4 Discrete-variable photonic processing

Photonic encodings use polarization, path, time bins, or other controlled modes. Linear optics transforms modes, while multiphoton interference depends on indistinguishability in all unobserved degrees of freedom. A polarization match alone is insufficient if timing or spectrum reveals which photon followed which path.

Probabilistic entangling operations can be useful when success is heralded, but source efficiency, coupling, propagation, detection, and feedforward determine usable rates. A failed detection can reflect loss or detector inefficiency, and a click need not guarantee the intended single-photon event. The protocol model should retain these distinctions. Loss-tolerant encodings and multiplexing trade additional modes and hardware for improved success, making mode count and detector resources as relevant as abstract qubit count.

Detailed study scope. Develop polarization, path, and time-bin encodings, interference of indistinguishable photons, probabilistic entangling operations, heralding, and detector models. Count source brightness, loss, mode matching, and feedforward in a photonic protocol.

11.5 Gaussian states and operations

For quadratures $R = (q_1, p_1, \dots)$ with commutator matrix Ω , define the covariance $V_{jk} = \langle \{R_j - d_j, R_k - d_k\} \rangle / 2$. Physical covariances satisfy $V + i\Omega/2 \geq 0$. A Gaussian state is fully specified by its displacement d and covariance V , providing an efficient description of a restricted but important optical family.

Gaussian unitaries act through symplectic transformations S , giving $V \mapsto SVS^T$ and corresponding displacement changes. A Gaussian channel acts as $V \mapsto XVX^T + Y$, with noise constrained by complete positivity. Squeezing redistributes quadrature uncertainty, while loss adds environmental noise. These new phase-space tools use the oscillator from Foundations without rederiving its eigenstates or energies.

Detailed study scope. Introduce phase-space characteristic functions, covariance matrices, symplectic transformations, squeezing, and Gaussian channels. Use dimensionless quadratures with vacuum covariance $I/2$; refer to Foundations for the oscillator rather than repeating its spectrum.

11.6 Non-Gaussian resources and optical measurements

Homodyne detection estimates a selected quadrature relative to a phase reference; heterodyne detection jointly samples both quadratures with added measurement noise. Photon counting resolves a different observable structure and can support non-Gaussian state preparation or conditioning. The detector choice therefore determines both accessible information and backaction.

Gaussian states, Gaussian operations, and Gaussian measurements form an efficiently describable restricted family and cannot provide every desired information-processing capability. Non-Gaussian resources extend that family but introduce preparation and detection costs. Finite squeezing gives finite-width distributions rather than ideal phase-space points. Loss and detector inefficiency further broaden or bias the effective resource, so continuous-variable protocol analysis must carry the actual covariance and non-Gaussian features relevant to its task.

Detailed study scope. Develop homodyne and heterodyne detection, photon counting, non-Gaussian preparation, and the limitations of Gaussian-only processing. Connect finite squeezing and loss to continuous-variable communication and the bosonic encodings of Chapter 13.

11.7 Interfaces and integrated systems [advanced]

An interface converts or transfers information between distinct carriers, frequencies, or modules. Conversion efficiency measures survival, while added noise and mode fidelity determine whether the surviving signal still carries useful quantum information. A high-efficiency classical transducer need not preserve entanglement.

Integrated systems also require synchronization, memory loading, routing, and classical feedforward. A slow heralding message can consume the memory advantage of an otherwise

good link. Modular designs trade local connectivity for inter-module communication and scheduling costs. Build the system estimate by composing independently validated operations under compatible conventions, then test where correlations or shared control invalidate that composition. Component quality is necessary evidence, but end-to-end performance is the application-facing result.

Detailed study scope. Analyze transduction, optical links, memory loading, synchronization, classical feedforward, and modular architectures. Build a requirement matrix for a network or protected processor and distinguish component demonstrations from integrated system performance.

Example 11.1 · Loss acting on a squeezed mode

For vacuum covariance $I/2$, pure loss of transmissivity η acts as $V_{\text{out}} = \eta V_{\text{in}} + (1 - \eta)I/2$. With $V_{\text{in}} = \text{diag}(1/4, 1)$ and $\eta = 1/2$, the output is $\text{diag}(3/8, 3/4)$. Its determinant is $9/32 > 1/4$, so the initially pure Gaussian state becomes mixed.

Common pitfall

Gate fidelity, loss probability, logical failure rate, and delivered entanglement rate describe different tasks. A platform comparison must hold the task and resource assumptions fixed.

11.8 Exercises

Exercise 11.1

Analytical. Apply the same pure-loss channel with $\eta = 0.8$ to the input covariance in the example; find its determinant.

Exercise 11.2

Conceptual. Explain why indistinguishability matters in an interference-based photonic operation.

Exercise 11.3

Diagnostic. Diagnose a network interface claim that reports conversion efficiency but omits added noise.

Exercise 11.4

Computational. Propagate one- and two-mode covariance matrices through loss and compare physicality and squeezing.

The analytical exercise is discussed in Appendix A.

11.9 Laboratory: Optical loss and interface requirements

Propagate a squeezed Gaussian mode through a variable-loss channel and optionally a two-mode resource through unequal losses. Verify the covariance uncertainty condition and the identity and complete-loss limits. Deliver squeezing and purity versus loss, then construct a platform-neutral interface budget including detection, coupling, added noise, and memory time. The task concerns channel performance rather than oscillator diagonalization.

Use the submission and validation contract in Appendix C.

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12 Quantum error correction and decoding

Learning objectives

- Explain correctable errors through code conditions and syndromes.
- Construct stabilizer codes and analyze their logical operations.
- Evaluate decoding under a declared noise and measurement model.

Prior QIS chapters: 1, 4, 7.

Foundations reference

Pauli operators, commutators, quantum registers, and controlled operations. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 5.5, 5.6, 7.1, 8.2. These prerequisite topics are not redeveloped here.

12.1 From classical codes to quantum protection

A classical code separates valid strings so that a bounded number of flips can be diagnosed. Quantum protection must additionally preserve superpositions and relative phase, which rules

out simply copying an unknown state for majority voting. Encoding distributes logical information into a subspace where selected errors move it into distinguishable error sectors.

Syndrome measurements identify these sectors without identifying the logical amplitudes. Bit-flip and phase-flip protection must both be considered, and arbitrary single-qubit errors can be expanded in a Pauli basis. Correcting a spanning error set then protects coherent combinations under the correctability conditions. This connects classical parity-check intuition to quantum recovery while making clear why the three-bit repetition example alone is not a complete quantum code.

Detailed study scope. Introduce parity checks, distance, redundancy, and decoding as specialized coding tools. Explain why quantum protection must preserve coherence and handle both bit and phase errors while syndrome extraction avoids revealing the encoded logical information.

12.2 Correctability and recovery

Let P project onto the code space and let E_a denote the error set. The Knill–Laflamme condition is $PE_a^\dagger E_b P = c_{ab}P$. It says that error overlaps do not reveal which logical state was encoded, allowing a recovery operation to restore all logical superpositions.

If an environment can distinguish logical basis states through the error process, coherence is lost in a way that syndrome correction cannot generally undo. The condition therefore expresses information isolation as well as algebraic correctability. Recovery performance can be tested by preserving entanglement with a reference, not merely by restoring two basis states. Approximate correction relaxes exact equality and requires a quantitative performance criterion tied to the intended channel or memory task.

Detailed study scope. Develop the Knill–Laflamme conditions, error spaces, recovery maps, and entanglement fidelity. Distinguish correcting an error set from correcting an arbitrary physical process and relate approximate correction to operational performance.

12.3 Stabilizer and CSS codes

A stabilizer code is the common $+1$ eigenspace of an abelian Pauli subgroup not containing $-I$. With $n - k$ independent generators on n qubits, the code space has dimension 2^k . Logical Paulis commute with the stabilizer while acting nontrivially on the code; distance is the minimum weight of such an undetectable nontrivial operator.

Binary symplectic representation converts Pauli commutation into a parity calculation and supports efficient syndrome tracking. CSS codes separate suitable X - and Z -type checks linked to classical codes. Repetition, five-qubit, and Steane examples illustrate different protection capabilities. Code parameters must be verified from independent generators and logical operators, rather than inferred from the number of drawn check symbols alone.

Detailed study scope. Develop Pauli groups, commuting generators, binary symplectic representation, logical operators, and CSS construction. Use repetition, five-qubit, and Steane codes as progressively richer examples with explicit limits and parameter checks.

12.4 Syndrome extraction and noisy measurements

Syndrome extraction couples data to ancillas and measures check information. A fault on an ancilla can propagate through later interactions to multiple data qubits, so the extraction circuit matters as much as the abstract stabilizer. Ordering, verification, flags, and repeated measurements can constrain harmful propagation.

With noisy readout, a single inconsistent check may be a measurement error rather than a data error. Repeated rounds produce a spacetime record from which a decoder infers a likely history. Leakage can persist across rounds and create correlated

faults outside a simple Pauli model. A benchmark should distinguish ideal-syndrome recovery, phenomenological data-and-measurement noise, and full circuit-level noise. Results from one model do not automatically certify performance in another. **Detailed study scope.** Construct syndrome circuits with ancillary checks and analyze error propagation, repeated measurements, leakage, and correlated faults. Distinguish ideal projective syndrome information from the output of a noisy extraction circuit.

12.5 Surface codes and topological structure

Surface-code checks are local on a lattice, and logical operators follow nontrivial paths connecting appropriate boundaries or winding around the topology. Distance measures the shortest harmful path. A sparse syndrome identifies endpoints or changes caused by error chains; many chains can share the same syndrome but differ in logical effect.

Measurement errors extend the decoding problem through time, so detection events are interpreted in a spacetime graph. Boundaries, defects, and code deformations alter the allowed paths. Local checks are attractive for constrained connectivity, but overhead depends on distance, cycle design, and the requested logical operations. A single lattice drawing is therefore the beginning of an architecture model, not a complete estimate of protected computation cost.

Detailed study scope. Develop lattice checks, boundaries, logical strings, distance, and spacetime syndrome records. Connect code geometry to locality and logical failure mechanisms without treating a single syndrome round as sufficient under measurement noise.

12.6 Decoding and performance

A decoder maps the observed syndrome to a recovery or logical-frame update. Maximum-likelihood decoding selects a likely logical class under the noise model, while practical algorithms

approximate this inference. Minimum-weight matching is effective for suitable graph-like error models; belief propagation exploits sparse factorization but can face degeneracy and short-cycle difficulties.

Performance should be reported as a logical failure probability per defined operation or cycle, with uncertainty and finite-size data. Decoder speed and latency matter when corrections influence subsequent operations. A decoder tuned to independent noise may perform differently under bias or correlations. Comparing decoders requires identical syndrome information and a common noise ensemble, including whether measurement faults, leakage, and correlated gates are present.

Detailed study scope. Compare minimum-weight matching, belief propagation, and small-instance maximum-likelihood decoding in their appropriate settings. Report logical error by event and cycle, finite-size behavior, decoder latency, and mismatch between simulated and physical noise.

12.7 Subsystem and sparse-check codes [advanced]

Subsystem codes introduce gauge degrees of freedom, permitting measurement of simpler gauge operators while protecting a logical subsystem. Gauge fixing changes which degrees of freedom are constrained and can enable different logical operations. The price and benefit depend on measurement schedules and error propagation, not just on lower check weight.

Quantum low-density parity-check codes keep check weights and qubit participation bounded while seeking favorable rate and distance. Their mathematical parameters can be attractive, but implementing long-range connections or complex extraction may be costly on a local device. Advanced comparison should connect code families to realistic connectivity, decoding, and logical gates. A theorem about code parameters does not by itself establish a complete fault-tolerant architecture.

Detailed study scope. Introduce gauge degrees of freedom, subsystem codes, quantum low-density parity-check families, and

rate-distance-connectivity tradeoffs. State what the code construction establishes and what still depends on extraction circuits, decoding, and architecture.

Example 12.1 · A limited but instructive recovery

The three-bit repetition code corrects one independent bit flip when the syndrome and recovery are perfect. Its logical failure probability is $3p^2(1-p)+p^3 = 3p^2-2p^3$. At $p = 0.1$, it is 0.028. The code does not correct arbitrary phase errors, so this number is not a general quantum-memory error rate.

Common pitfall

Distance alone does not determine an implemented logical error rate. The noise model, extraction circuit, decoder, and temporal correlations also matter.

12.8 Exercises

Exercise 12.1

Analytical. Compute the repetition-code failure probability at $p = 0.05$ and compare it with the uncoded bit-flip probability.

Exercise 12.2

Conceptual. Explain how a syndrome reveals error information without measuring the logical amplitudes.

Exercise 12.3

Diagnostic. Diagnose a code benchmark that assumes perfect measurement while claiming circuit-level performance.

Exercise 12.4

Computational. Enumerate error patterns for a small stabilizer code and verify syndrome and logical-class assignments.

The analytical exercise is discussed in Appendix A.

12.9 Laboratory: Syndromes and logical failures

Implement a small stabilizer code and a table decoder for an explicitly bounded error set. Exhaustively verify noiseless logical preservation and correction of the declared single-qubit errors. Estimate logical failure under sampled Pauli noise, then introduce syndrome-readout errors and repeated rounds. Deliver separate data-noise and measurement-noise results, uncertainty intervals, and examples of decoder failures.

Use the submission and validation contract in Appendix C.

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13 Fault tolerance and advanced encodings

Learning objectives

- Translate error correction into fault-tolerant computation.
- Budget logical operations and non-Clifford resources.
- Compare alternative encodings under finite physical resources.

Prior QIS chapters: 7, 11, 12.

Foundations reference

Circuit composition and oscillator states. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 8.3, 11.3. These prerequisite topics are not redeveloped here.

13.1 Fault propagation and threshold assumptions

Error correction becomes fault tolerance only when preparation, gates, syndrome extraction, recovery, and measurement remain reliable despite faults in those same operations. A gadget must prevent a small number of physical faults from becoming an uncorrectable logical error. Malignant combinations are fault sets that defeat this protection and determine the leading failure mechanisms.

Threshold theorems establish scalable suppression under specified local-noise and implementation assumptions when physical errors are sufficiently small. They do not provide one universal threshold for all devices or algorithms. Correlated noise, leakage, long-range faults, or slow decoding can challenge the model. The practical question is the logical error achieved at finite overhead, together with the evidence that the assumed fault model describes the implementation.

Detailed study scope. Define fault-tolerant gadgets, malignant fault sets at an introductory level, and threshold-theorem assumptions. Distinguish a threshold from a device specification and examine locality, correlation, leakage, and decoding-time requirements.

13.2 Logical Clifford operations

A transversal operation acts separately across corresponding physical qubits and limits error propagation within a block. Some logical gates admit this form for a chosen code, but a universal protected gate set generally requires additional mechanisms. In surface-code architectures, lattice surgery measures joint logical information by changing check boundaries; code deformation changes the encoded geometry over time.

These procedures consume repeated syndrome rounds and may require routing space, ancillas, and classical feedforward. A logical Clifford operation that is simple algebraically can therefore have a nontrivial spacetime cost. Tracking a Pauli frame can avoid immediate physical corrections, but later non-Clifford operations or measurements may depend on that record. Resource estimates must follow the implemented schedule rather than count abstract logical symbols alone.

Detailed study scope. Develop transversal gates, code deformation, lattice surgery, state preparation, and logical measurement. Count spacetime volume and routing requirements instead of equating one logical gate with one physical operation.

13.3 Non-Clifford resources

Stabilizer operations are not a universal computational resource on their own. Gate injection consumes a prepared non-stabilizer state to implement a non-Clifford operation through measurements and feedforward. Distillation uses noisy resource states and protected processing to produce fewer states of higher quality, provided input noise and circuit faults lie in an appropriate regime.

A factory is the scheduled collection of these preparation and distillation operations. Its throughput must match the algorithm's non-Clifford demand, while its output error must fit the total failure budget. More parallel factories reduce waiting but increase hardware. The resource-theory idea of magic thus becomes a countable operational expense: raw-state production, distillation levels, routing, storage, acceptance probability, and consumption rate all enter the architecture.

Detailed study scope. Introduce gate injection, magic-state distillation, factory scheduling, and non-Clifford gate synthesis. Relate the resource theory from Chapter 4 to practical magic-state consumption and explain why universal transversal protection faces restrictions.

13.4 Logical resource estimation

Begin with logical width, a dependency graph of operations, a synthesis tolerance, and a permitted total failure probability. Allocate errors across logical locations, then choose code parameters and schedules predicted to satisfy those allocations under a declared physical model. A union bound can provide a conservative total guarantee without requiring independence, although the individual bounds still need justification.

Physical qubit count includes data blocks, syndrome ancillas, routing space, and factories. Runtime depends on code cycles, critical-path dependencies, factory throughput, and feedforward. Sensitivity analysis should vary physical error rates and model coefficients rather than present one precise-looking number. A resource estimate is conditional on its assumptions;

it is not evidence that a current device can execute the algorithm.

Detailed study scope. Convert an algorithm's logical width, depth, synthesis tolerance, and failure budget into a code-distance and factory estimate under a declared model. Separate logical data, routing, ancillas, and magic-state production; include parallelism and cycle time.

13.5 Bosonic and continuous-variable codes

Bosonic encodings store a logical system in an oscillator's larger state space. Cat and binomial codes exploit structured superpositions or occupation patterns to diagnose selected errors such as photon loss. GKP encodings use a phase-space grid so that small displacement errors can be inferred through modular quadrature information.

Ideal grid states require infinite energy and are mathematical limits. Finite squeezing introduces intrinsic shift uncertainty, while loss, measurement noise, and imperfect ancilla interactions affect recovery. An outer qubit code can protect residual logical errors from an inner bosonic code, but the interface and error correlations must be modeled. The new subject is encoding and recovery in phase space, using the Gaussian tools of Chapter 11 rather than repeating the harmonic oscillator spectrum.

Detailed study scope. Develop cat, binomial, and Gottesman–Kitaev–Preskill encodings, syndrome extraction, and dominant loss or displacement errors. Include finite-energy states, finite squeezing, leakage, and the interface between bosonic and outer qubit codes.

13.6 Topological computation and alternative protection [advanced]

Topological computation stores information in fusion spaces of suitable excitations. Braiding acts on that encoded space, and

its computational power depends on the anyon model and allowed measurements. Topological protection can suppress sensitivity to some local perturbations without making every preparation, braid, or readout perfect.

Decoherence-free subspaces and noiseless subsystems use symmetries of the system-environment coupling, while autonomous correction engineers dynamics that return errors toward a protected manifold. Each approach has a different source of protection and a different failure mode when its assumptions break. Advanced study should compare explicit encoded operations, noise sensitivity, measurement requirements, and universality, rather than grouping all passive or exotic encodings under one undifferentiated robustness claim.

Detailed study scope. Introduce anyonic fusion spaces, braiding, measurement-based topological operations, and universality constraints. Compare protected subspaces, decoherence-free subsystems, and autonomous correction without assuming all proposed mechanisms offer the same guarantees.

13.7 Architecture and end-to-end reliability

End-to-end reliability combines protected computation with waiting, communication, calibration, and classical decisions. A logical memory experiment tests storage under its cycle and noise conditions; it does not automatically validate a universal processor using the same code. Non-Clifford operations, leakage handling, and intermodule transfer can introduce additional bottlenecks.

Compare candidate architectures on the same logical workload and accuracy target. Retain correlations that survive across operations, and account for decoder latency where it affects control. A smaller physical footprint can require longer runtime and hence more memory exposure. The result should be a transparent tradeoff between failure probability, physical space, and execution time, with the domain of validity of each empirical or analytical model stated.

Detailed study scope. Compare code families and layouts using the same algorithm, error model, and target failure probability.

Account for correlated failures, memory during waiting, decoding latency, and calibration. Relate a logical demonstration to the resources needed for a complete computation.

Example 13.1 · A whole-computation failure budget

For $N = 10^6$ logical locations, a union bound gives $P(\text{at least one failure}) \leq \sum_i p_i \leq Np_L$. To keep this bound at or below 0.01, require $p_L \leq 10^{-8}$ at every location. Independence is not needed for the union bound; obtaining each marginal bound under correlated physical noise is a separate challenge.

Common pitfall

A below-threshold physical error rate does not by itself specify the size or runtime of a useful protected computer. The target algorithm and total failure budget are essential.

13.8 Exercises

Exercise 13.1

Analytical. For 2×10^7 logical locations and total budget 10^{-3} , find a uniform per-location bound.

Exercise 13.2

Conceptual. Explain why magic-state factories can dominate a resource estimate.

Exercise 13.3

Diagnostic. Diagnose an estimate that uses ideal infinitely squeezed GKP states as finite-energy hardware.

Exercise 13.4

Computational. Compare two code-distance choices using a supplied logical-error model, recording its validity range and factory overhead.

The analytical exercise is discussed in Appendix A.

13.9 Laboratory: A transparent fault-tolerant resource estimate

Use a supplied logical circuit and an explicitly labeled illustrative logical-error model to allocate failure budgets, select candidate distances, and estimate data and factory spacetime costs. Validate dimensional consistency and monotonicity when the target budget tightens. Sweep physical error and parallel factory count. Deliver the assumptions, sensitivity plots, and separate logical, physical, and runtime totals; do not present model outputs as measured hardware capability.

Use the submission and validation contract in Appendix C.

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14 Characterization, certification, and error mitigation

Learning objectives

- Choose characterization methods suited to the claim and data budget.
- Separate preparation, measurement, process, and drift errors.
- Evaluate mitigation by bias, variance, and resource cost.

Prior QIS chapters: 1, 2, 12.

Foundations reference

Finite sampling, the Bloch sphere, and density matrices. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 3.5, 6.3, 7.3. These prerequisite topics are not redeveloped here.

14.1 Finite-data estimation and physical constraints

Ideal tomography in Foundations assumes exact expectation values. Finite data can produce a linear estimate with negative

eigenvalues even when the experimental state is physical. Constrained estimation instead optimizes an objective over the positive, trace-one state set, often using a likelihood derived from the measurement counts.

Enforcing physicality can introduce estimator bias, so a physical-looking reconstruction is not automatically more accurate in every loss function. Confidence regions, bootstrap procedures with stated assumptions, and held-out model checks assess uncertainty and fit. Calibration error and drift may dominate sampling error. The central question is which state properties are identifiable from the selected measurements and how strongly the finite record supports the reported conclusion, not how to repeat ideal Bloch-vector reconstruction.

Detailed study scope. Develop constrained state estimation, likelihood objectives, identifiability, confidence regions, and model checking. Assume ideal Bloch tomography from Foundations and focus on finite data, nonphysical linear estimates, calibration uncertainty, and the cost of enforcing positivity.

14.2 Process and gate-set characterization

Process tomography reconstructs a channel from a spanning set of preparations and measurements, while ancilla-assisted tomography probes correlations with a reference. Both require trusted aspects of the experimental setup. If preparation and measurement are uncertain, errors can be wrongly attributed to the process itself.

Gate-set tomography estimates a self-consistent family of states, gates, and effects from sequence data. Simultaneous transformations can leave all predicted probabilities unchanged, producing a gauge freedom in the representation. Gauge-dependent matrix entries should not be reported as uniquely measured physical properties. Useful outputs include identifiable predictions, model adequacy, and operational quantities with a clear gauge treatment. The chosen sequence set and stationarity assumptions determine what can actually be inferred.

Detailed study scope. Develop process tomography, ancilla-assisted methods, gate-set tomography, and gauge freedom.

Distinguish state-preparation-and-measurement errors from process errors and explain which quantities are identifiable without independently trusted devices.

14.3 Randomized benchmarking and drift

Randomized benchmarking averages over randomized gate sequences and fits a decay whose parameters summarize performance under a specified model. Randomization can simplify coherent structure, but gate dependence, leakage, drift, and temporal correlations can modify the interpretation. An exponential-looking curve alone does not establish every assumption behind a conversion to average error.

Interleaved variants compare reference sequences with sequences containing a target operation, while cycle benchmarking focuses on repeated operation layers. Leakage-sensitive measurements track population outside the intended space. A complete report states sequence sampling, length range, fitting uncertainty, model checks, and what the inferred parameter represents. Worst-case channel distance and algorithm-level success require additional reasoning beyond the fitted decay constant.

Detailed study scope. Introduce randomized benchmarking, interleaved variants, cycle benchmarking, and leakage-sensitive diagnostics. State noise and stationarity assumptions and explain why a decay parameter is not automatically a worst-case gate error.

14.4 Classical shadows and observable estimation

A classical shadow constructs an unbiased state-related estimator from randomized measurements and a known inverse measurement map. The goal is often to predict many observables without reconstructing every state parameter. The variance depends strongly on the observable and the measurement ensemble, captured by suitable shadow-norm bounds.

For local randomized measurements, low-weight observables can be economical while arbitrary global properties remain costly. Reusing a dataset for many predictions is valuable, but confidence guarantees must account for the number and choice of observables. Adaptive selection after inspecting the data can require additional analysis. The method therefore changes the measurement strategy for a specified family of questions; it does not turn a small record into unrestricted classical access to an unknown many-qubit state.

Detailed study scope. Develop randomized measurement estimators, shadow norms at an introductory level, and many-observable prediction. Analyze sample complexity, observable structure, data reuse, and the limits of predicting arbitrary global properties efficiently.

14.5 Certification and verification

Certification turns data into a bounded claim, such as minimum target fidelity or the presence of entanglement. Direct fidelity estimation can exploit target structure, while witnesses test selected inequalities. Verification of a computation may instead use a known answer, consistency checks, interactive structure, or a problem-specific certificate.

Sampling diagnostics such as cross-entropy-style scores probe particular features of an output distribution and rely on assumptions about reference probabilities and noise. A favorable score is not automatically a universal certificate of closeness or computational advantage. The report should identify the null hypothesis, trusted devices or calculations, statistical power, and failure modes the test cannot detect. This discipline keeps the strength of the conclusion matched to the evidence.

Detailed study scope. Compare entanglement witnesses, fidelity bounds, direct fidelity estimation, cross-entropy-style sampling diagnostics, and verified computations. Match the certification statement to trusted assumptions and separate a benchmark score from a general performance guarantee.

14.6 Readout mitigation and symmetry checks

Readout mitigation estimates a response matrix linking ideal outcome probabilities to observed frequencies and attempts to invert or regularize that map. An ill-conditioned response amplifies noise, and calibration uncertainty propagates into corrected estimates. Drift between calibration and measurement can create bias even when the inversion is implemented correctly.

Symmetry verification discards outcomes inconsistent with a known conserved property or code condition. This can reduce some errors but changes the accepted ensemble and consumes unsuccessful trials. Report acceptance probability and test whether postselection biases the target observable. A corrected quasi-probability vector need not itself describe a directly observed physical distribution. The appropriate comparison is estimator error at a fixed total experimental budget, including calibration and rejected events.

Detailed study scope. Develop calibrated response correction, conditioning, regularization, and symmetry verification. Include calibration uncertainty, drift, bias from postselection, and the fraction of accepted trials in any reported improvement.

14.7 Noise extrapolation and cancellation [advanced]

Zero-noise extrapolation fits or combines measurements at different effective noise levels to estimate a zero-noise limit. Its success depends on controllable noise scaling and a valid local model. Cancelling a leading bias often amplifies variance, as the two-point example shows; higher-order extrapolation can amplify it further.

Probabilistic error cancellation represents an inverse noisy process through signed weighted operations, trading bias reduction for sampling overhead. Virtual-distillation ideas use additional copies to emphasize dominant spectral components under their assumptions. These methods target expectation estimation and do not generally create a protected encoded computation. Evaluate residual model bias, variance, calibration cost, extra copies or circuit depth, and repeated-run mean squared error before declaring an improvement.

Detailed study scope. Develop zero-noise extrapolation, probabilistic error cancellation, and virtual-distillation ideas. Analyze model mismatch, sampling overhead, signed estimators, and bias-variance tradeoffs; distinguish mitigation of observables from quantum error correction of an encoded computation.

Example 14.1 · Extrapolation amplifies variance

For a linear-in-noise model, the two-point estimator $\hat{E}_0 = 2\hat{E}(\lambda) - \hat{E}(2\lambda)$ removes the leading bias. If the two independent estimates each have variance v , then $\text{Var}(\hat{E}_0) = 5v$. Standard error 0.02 at each noise level becomes about 0.0447 after extrapolation, before accounting for residual model bias.

Common pitfall

A mitigated mean closer to the exact value in one run does not establish lower total error. Compare bias, variance, calibration cost, and total shots over repeated runs.

14.8 Exercises

Exercise 14.1

Analytical. Find the extrapolated mean and standard error for independent estimates 0.70 ± 0.03 and 0.60 ± 0.03 , where the uncertainties are standard errors.

Exercise 14.2

Conceptual. Explain why gate-set tomography has a gauge freedom.

Exercise 14.3

Diagnostic. Diagnose a mitigation claim that excludes all rejected symmetry-check trials from its resource count.

Exercise 14.4

Computational. Compare raw and extrapolated estimators at the same total shot budget over repeated simulated experiments.

The analytical exercise is discussed in Appendix A.

14.9 Laboratory: Mitigation at a fixed measurement budget

Estimate a two-qubit circuit observable with tunable noise and compare raw measurement, readout correction, and two-point extrapolation. Include calibration shots in a fixed total budget and repeat the full experiment to estimate bias, variance, and mean squared error. Validate the noiseless reference and analytical variance propagation. Deliver a region where mitigation helps and, if present, a region where added variance outweighs reduced bias.

Use the submission and validation contract in Appendix C.

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PART V

Metrology and thermodynamics

15 Quantum sensing and metrology

Learning objectives

- Formulate parameter estimation with a complete physical resource budget.
- Compare classical and quantum Fisher information.
- Assess sensing gains under noise, nuisance parameters, and control constraints.

Prior QIS chapters: 1, 2, 3, 4.

Foundations reference

Sampling uncertainty, observable variance, uncertainty relations, and spin phase evolution. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 3.5, 6.2, 6.4, 10.3. These prerequisite topics are not redeveloped here.

15.1 Estimation tasks and identifiability

A sensing experiment specifies a parameter θ , a family of physical processes, an accessible measurement, and observed data. Identifiability asks whether different parameter values produce distinguishable distributions. A periodic phase signal can be locally sensitive yet globally ambiguous, so a narrow uncertainty interval requires prior localization or additional measurements.

Maximum-likelihood estimation selects parameters that make the observed record most probable. Bayesian estimation combines a likelihood with a stated prior and produces a posterior. Unknown contrast, offset, calibration, or drift creates nuisance parameters that must be fitted, bounded, or independently calibrated. A sensitivity number is meaningful only with this statistical model and a physical resource budget describing interrogation, preparation, readout, and repetition.

Detailed study scope. Specify the encoded parameter, signal model, preparation, interrogation, readout, and nuisance parameters. Introduce likelihood-based and Bayesian estimation, identifiability, local versus global ambiguity, and calibration requirements as specialized inference tools.

15.2 Fisher information and statistical bounds

Classical Fisher information is $\mathcal{F}(\theta) = \sum_x p(x | \theta) [\partial_\theta \log p(x | \theta)]^2$. Under regularity conditions, a locally unbiased estimator from ν independent trials satisfies $\text{Var}(\hat{\theta}) \geq 1/[\nu \mathcal{F}(\theta)]$. The derivation links estimator sensitivity to score variance through a Cauchy–Schwarz inequality.

The bound is local and does not guarantee an attainable finite-sample error for every estimator. Bias, parameter boundaries, multimodal likelihoods, and model mismatch can invalidate a naive comparison. Repeated simulated or experimental datasets can test whether an estimator reaches the expected regime. For correlated trials, information does not generally add as ν times a single-trial value, and the temporal model must be retained.

Detailed study scope. Develop classical Fisher information, the Cramér–Rao bound, bias conditions, repeated independent trials, and estimator attainability. Explain why a formal local bound need not describe a finite-data estimator near singular or ambiguous operating points.

15.3 Quantum Fisher information

The symmetric logarithmic derivative L_θ satisfies $\partial_\theta \rho = (L_\theta \rho + \rho L_\theta)/2$. Quantum Fisher information is $\mathcal{F}_Q = \text{tr}(\rho L_\theta^2)$ and bounds the Fisher information of allowed parameter-independent measurements in the standard local formulation. It describes an optimization over measurement choices, not necessarily the readout already installed in a sensor.

For a pure family generated by $e^{-i\theta G}$, $\mathcal{F}_Q = 4 \text{Var}(G)$. This connects probe preparation to sensitivity, but practical attainability also depends on operating-point knowledge and measurement access. In mixed noisy models, solving the logarithmic-derivative relation or using a spectral formula exposes information lost to decoherence. Units follow the encoded parameter and generator and should remain explicit.

Detailed study scope. Introduce the symmetric logarithmic derivative, optimization over measurements, pure-state generator variance, and the quantum Cramér–Rao bound. Separate the best allowed measurement from the measurement implemented in a particular sensor.

15.4 Interferometry under realistic noise

A Ramsey-type signal with visibility C has probability $[1 + C \cos \phi]/2$. The derivative is largest near quadrature, but sensitivity depends on both derivative and outcome variance. Substitution into Fisher information gives C^2 at quadrature. This is an estimation calculation using the elementary phase evolution already known from Foundations.

If contrast decays with interrogation time, longer exposure increases accumulated phase while reducing visibility. For a frequency signal $\phi = \omega t$, information about ω includes a factor t^2 , while the number of repetitions is limited by total time and dead time. Optimizing the full expression can favor a finite interrogation duration. Readout errors, unknown contrast, and reference noise further change the optimum and should be included before comparing sensors.

Detailed study scope. Use Ramsey and optical interferometry as estimation tasks with contrast loss, relaxation, imperfect

readout, and finite interrogation time. Refer to Foundations for elementary spin phase evolution; develop operating-point selection and total-time optimization here.

15.5 Entanglement and collective sensing

Independent probes typically give information proportional to particle number under matched single-probe resources. Entangled probes can increase the variance of a collective generator, producing quadratic information scaling in ideal noiseless phase-estimation models. Spin squeezing offers a different route by redistributing collective fluctuations relative to the measured signal.

These scaling statements require explicit assumptions about total particles, interrogation time, allowed preparation, and measurement. Local noise can destroy fragile large-scale coherence and change the asymptotic gain. Preparing entanglement may also consume time or probabilistic resources. The useful comparison therefore evaluates final estimator performance under the same total experimental budget and noise model, rather than comparing only the generator variance of ideal input states.

Detailed study scope. Analyze spin squeezing, GHZ-enhanced phase encoding, collective observables, and resource-normalized scaling. Compare standard and Heisenberg scaling under explicit noiseless assumptions, then examine the effect of local and correlated noise.

15.6 Adaptive and distributed sensing [advanced]

Adaptive sensing updates a control setting or measurement basis using earlier outcomes to keep the experiment informative. This can resolve ambiguity and approach a locally optimal readout, but feedback latency and estimation error affect performance. Bayesian and frequentist designs use different optimization criteria and should state which one is being pursued.

With several parameters, a Fisher-information matrix describes local correlations in estimation. Quantum-optimal measurements for different parameters can be incompatible, so separate single-parameter bounds may not be jointly attainable. Distributed sensors add network resources and may target a weighted global parameter rather than every local value. A claimed distributed gain must include entanglement distribution, synchronization, loss, and memory, linking metrology back to the network analysis in Chapter 5.

Detailed study scope. Develop adaptive phase estimation, nuisance-parameter strategies, multiparameter compatibility, and networked sensors. Relate distributed gains to the costs of generating and maintaining the shared resource from Chapter 5.

15.7 Applications and evidence

Magnetometry converts a field into an accessible spin response; clocks compare phase accumulation to a reference; imaging and spectroscopy estimate spatial or frequency-dependent structure. Each application adds priorities such as bandwidth, dynamic range, spatial resolution, and stability that are not captured by one local phase variance.

A narrow-band sensor with excellent sensitivity may miss rapidly varying signals, while high gain can reduce unambiguous range. Calibration drift and dead time can dominate long-duration performance. Report the target signal model, operating range, resource count, and uncertainty method together with sensitivity. The strongest evidence compares complete experimental cycles against a matched classical or unenhanced reference and explains which improvements originate from quantum resources, control, or ordinary statistical averaging.

Detailed study scope. Formulate magnetometry, clocks, imaging, and spectroscopy through their sensitivity, bandwidth, spatial resolution, and dynamic range. Include dead time, control overhead, reference noise, and calibration drift in a fair comparison with classical sensing strategies.

Example 15.1 · Contrast reduces phase information

For $p(0 | \phi) = [1 + C \cos \phi]/2$, the classical Fisher information at quadrature $\phi = \pi/2$ is C^2 . With contrast $C = 0.8$ and $\nu = 400$ independent trials, the local unbiased-estimator standard-deviation bound is $1/(C\sqrt{\nu}) = 0.0625$ radians. This is a statistical bound at the chosen operating point, not a guaranteed error in one experiment.

Common pitfall

A phase sensitivity advantage can disappear after counting preparation time, discarded trials, decoherence, calibration, and the resources used to create entanglement.

15.8 Exercises

Exercise 15.1

Analytical. At contrast $C = 0.5$ and $\nu = 900$, calculate the local standard-deviation bound at quadrature.

Exercise 15.2

Conceptual. Explain why quantum Fisher information need not be attained by a fixed laboratory readout.

Exercise 15.3

Diagnostic. Diagnose a Heisenberg-scaling claim that keeps particle number but changes total interrogation time.

Exercise 15.4

Computational. Compare a likelihood estimator with its local Fisher bound over repeated noisy phase-estimation experiments.

The analytical exercise is discussed in Appendix A.

15.9 Laboratory: Sensing with noise and time costs

Simulate a contrast-limited phase sensor and estimate phase by likelihood over repeated datasets. Validate empirical variance against Fisher information in an identifiable local regime, then expose phase ambiguity or small-sample deviations. Optimize interrogation time under a specified decay law and fixed total time including dead time. Deliver sensitivity versus time and a comparison with a matched unentangled or unenhanced reference.

Use the submission and validation contract in Appendix C.

References and further reading

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16 Quantum thermodynamics and information resources

Learning objectives

- Relate information processing to thermodynamic state transformations.
- Distinguish work, heat, entropy production, and memory costs.
- Evaluate finite-system and quantum-resource claims under explicit operations.

Prior QIS chapters: 1, 2, 3, 4.

Foundations reference

Probabilities, energy, Hamiltonian eigenstates, and conserved quantities. For further study, consult *Mathematical and Physical Foundations for Quantum Computing*, Sections 3.1, 9.1, 10.2, 10.5. These prerequisite topics are not redeveloped here.

16.1 Thermal states and operational thermodynamics

A thermal state is $\tau = e^{-\beta H}/Z$, where $Z = \text{tr}(e^{-\beta H})$ and $\beta = 1/(k_B T)$. This introduces equilibrium statistical mechanics as a specialized tool beyond the energy concepts in Foundations. A

thermodynamic task also needs a bath, a work-storage system, and a specification of allowed controls.

Passivity means that no work can be extracted from the state by a cyclic unitary on the system under the stated Hamiltonian. Thermal states have stronger stability properties under combining copies. Thermal operations impose energy conservation on system-bath interactions, while other free-operation choices define different resource theories. Average internal energy is therefore not synonymous with useful work: extractability depends on populations, coherence, available references, and the operation class.

Detailed study scope. Introduce Gibbs states, partition functions, passivity, work storage, baths, and thermal operations as new specialized tools. State the system-bath-control boundary and distinguish average energy from extractable work.

16.2 Free energy and irreversibility

Using dimensionless entropy $s(\rho) = -\text{tr}(\rho \ln \rho)$, nonequilibrium free energy is $F_T(\rho) = \text{tr}(H\rho) - k_B T s(\rho)$. For a fixed Hamiltonian, $F_T(\rho) - F_T(\tau) = k_B T D_{\ln}(\rho \| \tau)$, where relative entropy uses natural logarithms. The nonnegative excess quantifies a resource relative to equilibrium in the stated setting.

A Gibbs-preserving channel contracts this relative entropy, giving free-energy monotonicity. Heat, work, and entropy production require a specified system-control boundary; changing the Hamiltonian introduces work accounting beyond a fixed-channel relaxation step. Strong coupling and initial system-bath correlations can invalidate a simple weak-coupling decomposition. The equations are therefore useful constraints when their model applies, not universal bookkeeping rules independent of physical boundaries.

Detailed study scope. Develop nonequilibrium free energy, relative entropy to equilibrium, entropy production, and second-law inequalities under declared dynamics. Explain how coherence and strong coupling can require a more careful description than population-only thermodynamics.

16.3 Erasure and information engines

Resetting an unbiased degenerate memory reduces its entropy by $k_B \ln 2$. In an ideal quasistatic isothermal cycle, compensating this reduction requires at least $k_B T \ln 2$ of work and corresponding heat transfer to the bath. A biased memory has a different initial entropy, and side information can change the operational reset task.

A measurement-feedback engine uses an outcome to choose a favorable control action, but the outcome is stored in a physical memory. Completing a repeatable cycle requires accounting for that memory and any correlations or reference resources. Omitting reset can make a partial cycle appear to beat a full-cycle bound. Finite time, unreliable reset, and practical control dissipation alter achievable costs; the ideal lower bound is not a direct estimate of processor energy consumption.

Detailed study scope. Derive Landauer's bound for a degenerate classical memory reset and examine its quasistatic, bath, and reliability assumptions. Develop measurement-feedback cycles and account for the memory, controller, correlations, and reset cost across the complete cycle.

16.4 Work statistics and fluctuation relations

A two-point energy-measurement scheme defines a work distribution by measuring initial and final energies around a specified drive. For an isolated driven system initially in a Gibbs state, the Jarzynski relation takes the form $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$ under the standard protocol assumptions. Jensen's inequality then implies the average-work inequality $\langle W \rangle \geq \Delta F$.

Individual trajectories can have work below the average bound without violating it. Crooks-type relations compare forward and reverse protocols with compatible initial equilibrium conditions. In quantum systems, the initial energy measurement can remove coherence, so this work definition does not answer every coherent-process question. State the measurement protocol, equilibrium preparation, and reverse process before interpreting a fluctuation relation or testing it numerically.

Detailed study scope. Introduce two-point energy measurements, work distributions, Jarzynski and Crooks relations in their appropriate settings, and the role of initial equilibrium. Distinguish trajectory-level fluctuations from violations of average thermodynamic constraints.

16.5 Coherence, correlations, and thermal resources

Coherence between energy sectors is a resource when the allowed operations respect time-translation symmetry. A state can have useful populations and coherence, but extracting value from the coherence may require a phase reference. Gibbs-preserving maps and thermal operations need not permit identical transformations, so resource claims must name the operation class.

Correlations can change work extraction or erasure costs even when local states appear unchanged. Quantum side information connects these tasks to conditional entropy, but an operational protocol and assistance convention are necessary to interpret a negative conditional value. Catalysts can enable transformations while being returned under a specified condition; allowing small error or residual correlations changes the resource account. These distinctions prevent thermodynamic benefit from being inferred solely from the presence of entanglement or coherence.

Detailed study scope. Apply the resource-theory framework to coherence, asymmetry, correlations, and catalytic transformations. Compare unrestricted operations with energy-conserving or Gibbs-preserving models, and connect conditional entropy to tasks with quantum side information.

16.6 Finite-size transformations and machines [advanced]

Finite systems fluctuate, so average extractable work differs from work guaranteed with a chosen failure probability. Single-shot quantities retain that reliability requirement and do not collapse every task to an asymptotic free-energy difference. Finite baths also change temperature or become correlated, limiting the ideal infinite-reservoir approximation.

Thermomajorization characterizes certain population transformations under thermal constraints, with ordering weighted by Gibbs factors. Quantum batteries and small engines raise additional questions about charging power, extractable work, collective control, and the cost of interactions. A faster charging process can consume stronger controls or more correlated resources. Compare complete protocols at matched energy, control, and reliability budgets before interpreting a collective enhancement as an unconstrained thermodynamic advantage.

Detailed study scope. Develop single-shot work, finite baths, thermomajorization, quantum batteries, and small heat engines. State the resource accounting behind charging power, collective enhancements, and finite-time tradeoffs; avoid equating coherence with universal thermodynamic benefit.

16.7 Information processing as a physical cycle

A complete information-processing cycle includes preparation, computation, communication, measurement, feedback, storage, and reset. Each stage changes energy, entropy, correlations, or an external resource. A ledger should identify where work enters, where heat goes, what information is retained, and which auxiliaries return to their initial condition.

This viewpoint connects the book's operational tasks without adding another foundations review. Communication may consume entanglement; fault tolerance uses measurements and

resets; sensing uses references and feedback. Their thermodynamic costs depend on the physical implementation and cycle, not only on abstract gate count. The final analysis should distinguish mathematical lower bounds, conditional model predictions, and measured engineering dissipation, then explain which constraints matter for the proposed application.

Detailed study scope. Integrate preparation, computation, communication, measurement, feedback, and reset into a complete resource ledger. Relate thermodynamic bounds to engineering dissipation without treating a lower bound as a measured device energy or a prediction of achievable efficiency.

Example 16.1 · A thermodynamic lower bound for erasure

For one unbiased classical bit in a degenerate memory, ideal quasistatic reset against a bath at $T = 300$ K has minimum work $k_B T \ln 2 \simeq 2.871 \times 10^{-21}$ J, with corresponding heat transferred to the bath in the ideal model. Real control, finite-time operation, and reliability requirements can increase the cost.

Common pitfall

Landauer's bound is not the energy consumed by every logic gate or by a measured processor. Specify which memory is erased, its initial uncertainty, and the allowed cycle.

16.8 Exercises

Exercise 16.1

Analytical. Find the ideal minimum reset work for 10^6 independent unbiased bits at 300 K.

Exercise 16.2

Conceptual. Explain why extracting work with feedback requires accounting for the controller memory.

Exercise 16.3

Diagnostic. Diagnose an information-engine claim that omits reset and reports only the favorable part of a cycle.

Exercise 16.4

Computational. Compute free-energy changes for a finite-level system under a specified thermal map and check the appropriate monotonicity.

The analytical exercise is discussed in Appendix A.

16.9 Laboratory: Thermal maps and complete cycles

Model a finite-level system relaxing toward a Gibbs state and track energy, entropy, and nonequilibrium free energy with consistent logarithms and units. Validate the stationary Gibbs state and free-energy monotonicity for the specified thermal channel. Add a classical measurement-feedback-reset ledger and identify every resource transfer. Deliver a cycle balance and sensitivity to temperature or reset reliability; no device-efficiency claim is inferred.

Use the submission and validation contract in Appendix C.

References and further reading

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A Selected solutions and assessment guidance

The following solutions address the analytical exercise in each chapter. The conceptual, diagnostic, and computational problems are assessment tasks; a complete solution manual and executable laboratory notebooks are separate instructional artifacts. State assumptions and units before interpreting a number.

A.1 Chapter 1: analytical exercise

$P_{\text{err}} = (1 - 1/\sqrt{2})/2 \simeq 0.1464$, compared with 0.25. Diagonalizing the difference of the prior-weighted states determines the two decision subspaces; the improvement is approximately 0.1036.

A.2 Chapter 2: analytical exercise

$1/T_\phi = 1/80 - 1/120 = 1/240$ in inverse microseconds, hence $T_\phi = 240 \mu\text{s}$. The subtraction is model-dependent; it is not a universal identity for arbitrary decay envelopes.

A.3 Chapter 3: analytical exercise

At $p = 0.4$, $(C, Q) = (0.6, 0.2)$; at $p = 0.6$, $(C, Q) = (0.4, 0)$. Positive classical capacity can coexist with zero unassisted quantum capacity. These asymptotic rates are not deterministic finite-block transmission counts.

A.4 Chapter 4: analytical exercise

Only the eigenvalue $(1 - 3p)/4$ can be negative, so $\mathcal{N} = \max(0, (3p-1)/4)$. The values are 0.2 and zero. The CHSH threshold for this family is $p > 1/\sqrt{2}$, higher than its entanglement threshold.

A.5 Chapter 5: analytical exercise

$E[\max] = 20 - 1/0.19 \simeq 14.7368$ rounds, compared with ten for one link. The maximum of two waiting times is not the same as the mean of either one. Deadlines and unsuccessful swaps change the result.

A.6 Chapter 6: analytical exercise

The added sifted-key error rate is $f/4 = 0.05$, assuming the attacked fraction is independent of bases and detection, with no other noise. This relationship does not identify all possible attacks producing the same observed error.

A.7 Chapter 7: analytical exercise

The total is $20 + 21 = 41$ CNOTs, of which $21/41 \simeq 51.2\%$ implement routing. The increase over the original circuit is 105%. These percentages have different denominators and must not be interchanged.

A.8 Chapter 8: analytical exercise

Three bits give 101 and $1 + 2 + 4 = 7$ uses of U . Four bits give 1010 and $1 + 2 + 4 + 8 = 15$ uses. The additional zero bit changes cost even though the phase was already exactly representable.

A.9 Chapter 9: analytical exercise

$A^{-1}|b\rangle$ is proportional to $3|0\rangle + |1\rangle$, giving $(3|0\rangle + |1\rangle)/\sqrt{10}$ and probability $9/10$. The normalization and amplitude information differ from a direct classical list of solution entries.

A.10 Chapter 10: analytical exercise

The gradient is $(0.3 + 0.1)/2 = 0.2$; its variance is $(0.02^2 + 0.02^2)/4 = 0.0002$, giving standard error approximately 0.01414. This result assumes independent estimates and the applicable parameter-shift generator.

A.11 Chapter 11: analytical exercise

The output is $\text{diag}(0.3, 0.9)$, with determinant 0.27. It satisfies the one-mode uncertainty condition $\det V \geq 1/4$ under the stated quadrature convention. Attenuation adds vacuum noise as well as reducing the input signal.

A.12 Chapter 12: analytical exercise

$3(0.05)^2 - 2(0.05)^3 = 0.00725$, compared with 0.05 without encoding. The improvement factor is approximately 6.90 for this bit-flip-only model with perfect recovery; no phase-error protection follows from it.

A.13 Chapter 13: analytical exercise

The union-bound allocation gives $p_L \leq 10^{-3}/(2 \times 10^7) = 5 \times 10^{-11}$. This is a target bound, not a prediction of an existing device or an assertion that all errors are independent.

A.14 Chapter 14: analytical exercise

The extrapolated mean is $2(0.70) - 0.60 = 0.80$. Its standard error is $\sqrt{5}(0.03) \simeq 0.0671$. This calculation assumes independent samples and does not include nonlinear-noise bias or uncertainty in the noise-scaling procedure.

A.15 Chapter 15: analytical exercise

The bound is $1/(0.5\sqrt{900}) = 1/15 \simeq 0.0667$ radians, under the same local unbiased-estimation and independent-trial conditions. Unknown contrast or phase ambiguity requires an enlarged estimation model.

A.16 Chapter 16: analytical exercise

The bound is $10^6 k_B T \ln 2 \simeq 2.871 \times 10^{-15}$ J. Additivity here assumes independent unbiased bits and the stated ideal erasure model. It excludes the practical costs of storage, control, and finite-speed operation.

A.17 Assessment guidance

Conceptual answers should explain a mechanism and its assumptions. Diagnostic answers should identify the invalid inference, give a counterexample or missing condition, and state a corrected claim. Computational submissions should report a reproducible method, an independent validation result, uncertainty or numerical convergence, and a resource account. The integrated project rubric appears in Appendix [D](#).

B Essential notation and conventions

The foundations volume supplies basic vector, operator, and Hamiltonian notation. This appendix fixes the additional conventions needed to prevent ambiguity in information-theoretic and computational statements.

SYMBOL	MEANING
ρ_A, ρ_{AB}	Notation as in Foundations, Sections 7.3–7.4; consult that volume for further study.
$\mathcal{E}$	A quantum channel: a completely positive trace-preserving linear map.
K_a	Kraus operators with $\sum_a K_a^\dagger K_a = I$ for a channel.
E_x	POVM effect; $p(x) = \text{tr}(\rho E_x)$.
$\mathcal{J}_x$	Outcome-conditioned quantum instrument map. Its trace gives the outcome probability.
$J(\mathcal{E})$	Unnormalized Choi operator $(\text{id} \otimes \mathcal{E})(\Omega\rangle\langle\Omega)$, where $ \Omega\rangle = \sum_i i\rangle i\rangle$.
$\text{tr}_{\text{out}} J$	Equals I_{in} for a trace-preserving map with the stated Choi convention.
$T(\rho, \sigma)$	Trace distance $\frac{1}{2}\ \rho - \sigma\ _1$.
$F(\rho, \sigma)$	Squared Uhlmann fidelity $(\text{tr} \sqrt{\sqrt{\rho}\sigma\sqrt{\rho}})^2$.
$S(\rho)$	Von Neumann entropy $-\text{tr}(\rho \log_2 \rho)$ in bits.

SYMBOL	MEANING
$h_2(p)$	Binary entropy $-p \log_2 p - (1 - p) \log_2 (1 - p)$, with $0 \log 0 = 0$.
$S(A B)$	Quantum conditional entropy $S(AB) - S(B)$, which may be negative.
$I(A : B)$	Mutual information $S(A) + S(B) - S(AB)$.
$D(\rho\ \sigma)$	Relative entropy $\text{tr}[\rho(\log_2 \rho - \log_2 \sigma)]$, infinite if the support condition fails.
ρ^{T_B}	Partial transpose on subsystem B in a declared basis.
$\mathcal{N}(\rho)$	Negativity $(\ \rho^{T_B}\ _1 - 1)/2$.
$[[n, k, d]]$	Code with n physical qubits, k logical qubits, and distance d .
p, p_L	Declared physical and logical error probabilities; the event and time interval must be specified.
T_1, T_2, T_ϕ	Relaxation, coherence, and pure-dephasing time in the stated phenomenological model.
q, p (bosonic)	Dimensionless quadratures with $[q, p] = i$. Here p is not an error probability.
V	Symmetrized quadrature covariance; the vacuum is $I/2$.
η	Optical transmissivity in the pure-loss example.
θ, ϕ	Rotation or phase parameter; units and the 2π convention are stated locally.
$\nu, \mathcal{F}$	Number of independent trials and Fisher information; classical or quantum type is specified.
ϵ	A tolerance, approximation error, or failure budget whose operational definition is given locally.

B.1 Register order and units

Unless an experiment declares otherwise, computational basis states are written $|q_0 q_1 \cdots q_{n-1}\rangle$ with q_0 the most significant bit. SDK display strings must be decoded using the backend's documented convention. A qubit permutation should be tracked through routing and measurement.

Information entropies use logarithms to base two. Thermodynamic formulas use natural logarithms unless explicitly converted, with entropy in J/K when multiplied by Boltzmann's constant. Frequencies in Hz and angular frequencies in rad/s differ by 2π . GiB and MiB denote binary memory units. Raw array size excludes workspaces, copies, software overhead, and operating-system memory.

B.2 Error language

State distance, average infidelity, logical failure probability, estimator variance, systematic bias, and runtime are different quantities. An error budget should never add incomparable quantities without a derivation that places them in the same operational guarantee. Statistical intervals should state their construction and the repeated-sampling or posterior interpretation.

C Teaching routes and laboratory practice

C.1 A complete sequence

The sixteen chapters form a book, not sixteen independent courses. The following proposed sequence allows time for derivations, computational work, and advanced discussion. It assumes Foundations has already been completed.

COURSE	CHAPTERS	FOCUS
I	1–4	Operations, open dynamics, information theory, and quantum resources.
II	5–8	Communication, security, computational models, and algorithm families.
III	9–12	Simulation, hybrid methods, physical architectures, and error correction.
IV	13–16	Fault tolerance, experimental evidence, sensing, and thermodynamics.

Each course contains sixteen weeks with four contact and eight independent hours per week: 64 contact plus 128 independent hours. The complete route is 256 contact plus 512 independent hours, or 768 learning hours. These are instructional planning estimates, not a claim of measured student workload.

Within each course, allocate 32 contact hours to exposition and derivations, 16 to the four chapter laboratories, eight to problem seminars, and eight to integrated project discussion and assessment. The laboratory meeting is an initial four-hour guided session; preparation, extensions, and reports use independent study time. Chapters with advanced material can require more attention than their nominal four-week share.

C.2 Dependencies and shorter routes

The numerical chapter sequence respects all declared prior QIS dependencies. An information-theory specialization can emphasize Chapters 1, 3–6, and 16 with Chapter 2 added for dynamics. A computation specialization uses Chapters 1, 7–10, and 12–14, adding Chapters 2, 4, 5, and 11 where their control, entanglement, network, or physical tools are needed. A sensing route uses Chapters 1–4 and 15, with Chapters 5 and 11 for distributed or optical extensions. These are reading emphases, not permission to omit required tools.

A one-semester survey should select core sections across the five parts, limit laboratories to a few representative tasks, and defer the marked advanced sections. It does not offer the same depth as the complete sequence. Instructors should publish the exact section selection and assessment scope before the course begins.

C.3 Laboratory environment and submission contract

Environment setup and basic reproducibility are assumed from Foundations, Chapter 14, Section 14.1. For further study of those procedures, return to that section. The new activities specify their QIS problem, method, validation, and deliverables within Chapters 1–16 instead of repeating setup lessons.

A laptop with 16 GiB RAM and ordinary double-precision numerical tools is sufficient for the specified baseline instances. No GPU or quantum processor is required. NumPy, SciPy, and

plotting tools support the common numerical layer; a specialized package is optional when its role and actual working version are recorded. Use the small qubit, mode, and block limits stated in each activity before attempting extensions.

Every submission includes the task and hypothesis, physical and statistical model, input data, recorded environment and random seeds, conventions, method, raw results, independent validation, uncertainty or convergence, resource account, and interpretation. Deterministic residual tolerances apply to numerical identities; sampling results require intervals or repeated runs. Do not force a stochastic estimate to match an exact number to an arbitrary decimal tolerance.

The specifications are designed instructional activities. Their inclusion does not imply that the corresponding notebooks were implemented or run as part of preparing this manuscript.

C.4 Assessment

A suggested course assessment allocates 25 percent to analytical and conceptual problems, 30 percent to laboratories, 30 percent to the integrated project, and 15 percent to oral or written synthesis. A strong answer states the task, justifies the model, develops the argument, tests the result, and explains its limits. The capstone rubric in Appendix D makes these expectations concrete for work connecting several chapters.

D Integrated projects and synthesis

These projects connect the sixteen thematic chapters without adding a seventeenth main chapter. Each has a bounded baseline suitable for classical simulation and an optional advanced extension. Select one project per course or develop one progressively across the sequence.

D.1 Communication with a security and memory budget

Use Chapters 1, 3–6, and 11 to compare a direct link with a two-link repeater for delivering a specified entanglement resource. Model loss, heralding, waiting time, memory decay, and classical coordination. Define whether the application is teleportation, trusted-node key relay, or an end-to-end security protocol; these require different claims.

Deliver a rate-fidelity-latency comparison, a trust diagram, and a complete resource ledger. Validate the ideal waiting-time limit and an independent small-instance channel calculation. If reporting a key length, identify the security theorem and every assumed finite-key term; otherwise label the security budget as an illustrative model. The advanced extension optimizes a discard or multiplexing policy under fixed memory resources.

D.2 An algorithm from access model to logical resources

Choose phase estimation, amplitude estimation, or a small scientific subroutine from Chapters 7–10. Define the requested output and precision, construct the input and access operations, and count coherent queries, ancillas, synthesis, and measurements. Compare with a classical method that answers the same question under comparable access assumptions.

Using Chapters 12–13, translate a supplied logical circuit into a conditional fault-tolerant estimate. Deliver separate tables for algorithmic and physical costs, a correctness check on small inputs, and a sensitivity analysis for precision and logical failure budget. The advanced extension examines factory throughput or a better input representation. A resource estimate must not be reported as a measured execution or a proof of practical advantage.

D.3 A protected memory with imperfect syndrome data

Use Chapters 1–2 and 12–14 to compare an unencoded memory with a small encoded memory under a declared noise model. Begin with ideal syndrome extraction, then include noisy measurements and repeated rounds. Specify whether the model is phenomenological or circuit-level and identify its correctable error set.

Deliver logical-failure curves with uncertainty, exhaustive checks for the small declared error set, and examples of harmful propagation or decoder ambiguity. Include decoder runtime or latency if it affects the protocol. The advanced extension compares a biased-noise model, another decoder, or a bosonic inner encoding with clearly stated additional assumptions.

D.4 Scientific simulation with a controlled error budget

Use Chapters 7–11 and 14 to simulate a noncommuting many-body model or a small fermionic Hamiltonian. Specify the observable and scientific accuracy, then separate model truncation, state preparation, approximation, sampling, and hardware errors. Compare product formulas or a variational method with an independent exact result in the tractable regime.

Deliver convergence plots, a matched classical comparison, and a total resource estimate for the requested observable. Explain which small-instance checks support an extension and which scaling questions remain open. The advanced extension studies entanglement growth, measurement allocation, or an alternative mapping. Repeating the elementary Ising spectrum is not an adequate QIS project.

D.5 Sensing under total-time and network constraints

Use Chapters 1–4 and 15 to estimate a phase or frequency with noisy contrast, calibration uncertainty, and a fixed total time. Compare the implemented estimator with its classical Fisher bound and the appropriate quantum bound. Include preparation, readout, and dead time in the repetition budget.

Deliver repeated-run estimation results, a test of local identifiability, and an optimized interrogation-time comparison against a matched reference. An optional network extension uses Chapters 5 and 11 to account for distributing an entangled probe. The final claim must distinguish a local statistical bound, a simulated estimator performance, and a physically demonstrated gain.

D.6 An information-processing thermodynamic cycle

Use Chapters 1–4 and 16 to model preparation, measurement, feedback, and reset in a finite system with a specified bath and work-storage convention. Identify changes in energy, entropy, correlations, and stored information throughout the cycle. Use free-energy monotonicity or a fluctuation relation only where its assumptions hold.

Deliver a complete cycle ledger, a stationary-state check, and a comparison with an ideal erasure bound. Vary bath temperature, memory bias, or reset reliability to expose which resource changes the result. The advanced extension includes quantum side information or a finite bath. Do not infer actual processor dissipation from an ideal lower bound.

D.7 Common project rubric

CRITERION	EVIDENCE	WEIGHT
Task and assumptions	Clear input, output, success criterion, model, and allowed resources.	20%
Technical reasoning	Correct derivations, justified methods, and explicit validity conditions.	25%
Validation	Independent checks, convergence or uncertainty, and appropriate comparisons.	25%
Resource accounting	Complete costs, overheads, and sensitivity to consequential assumptions.	20%
Communication	Reproducible record, readable figures, and conclusions matched to evidence.	10%

D.8 Synthesis questions

Explain how information lost to an environment connects channel capacity, error correction, and thermodynamic irreversibility. Then compare the roles of measurement in communication, protected computation, and sensing: what information must be revealed, preserved, or deliberately erased in each task? A useful synthesis identifies both the shared tools and the different success criteria, without collapsing every form of error into one number.

Recommended bibliography for further study

The chapter reading lists connect this original instructional synthesis to established texts, primary research, and author lecture notes. These sources provide full proofs and specialized treatments beyond the developed curriculum. Hardware papers supply physical principles and historical examples; they are not presented as current platform specifications.

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This syllabus edition continues the educational work of CyberColombia and the Quantum Colombia Alliance. It reuses the editorial and visual system of the English foundations book so that the two volumes can serve as a coherent learning sequence.

The cited authors and research groups provide the scientific literature supporting the developed curriculum and its further study. Inclusion in the bibliography does not imply their endorsement of this curriculum. The design also anticipates review by instructors in information theory, computation, communication, physics, and engineering as the manuscript develops into a complete proof-level textbook.

Contributing to the educational series

Suggestions about scope, prerequisite dependencies, examples, laboratory validation, or terminology can be directed to quantum@cybercolombia.org.

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